

ECON 2020: Principles of Macroeconomics

Lecture Notes

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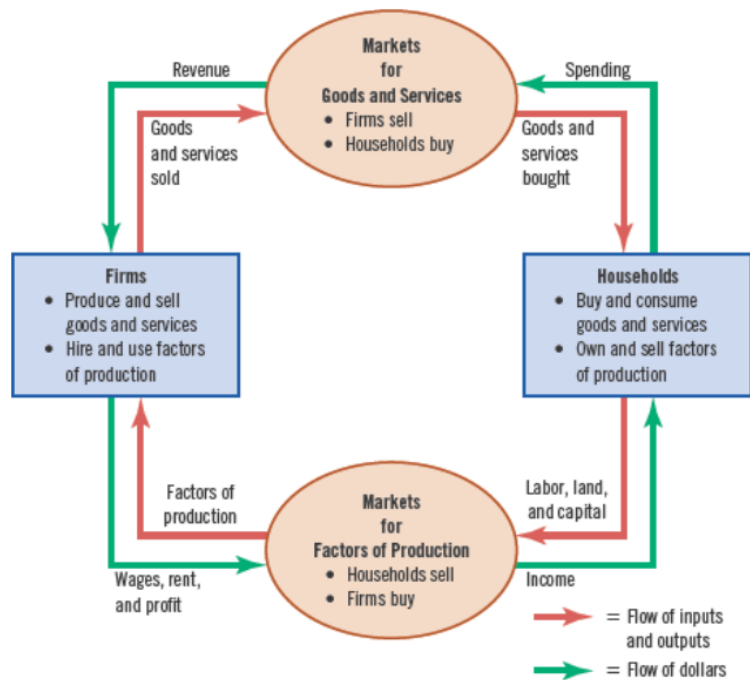
Chapter 1. Ten Principles of Economics

- How people make decisions
 - (1) People and Institutions face trade-offs
 - (2) The cost of Something is What you give up to get it
 - Opportunity cost: the cost of something is the value of the best alternative you give up to get it
 - (3) Rational People Think at the margin
 - (4) People respond to incentive
- How people interact
 - (5) Trade can make everyone better off
 - (6) Markets are usually a good way to organize economic activity
 - (7) Governments can sometimes improve market outcomes
- How the economy as a whole works
 - (8) A country's standard of living depends on its ability to produce goods and services
 - (9) Prices rise when the government prints too much money
 - (10) Society faces a short-run trade-off between inflation and unemployment

Chapter 2. Thinking Like an Economist

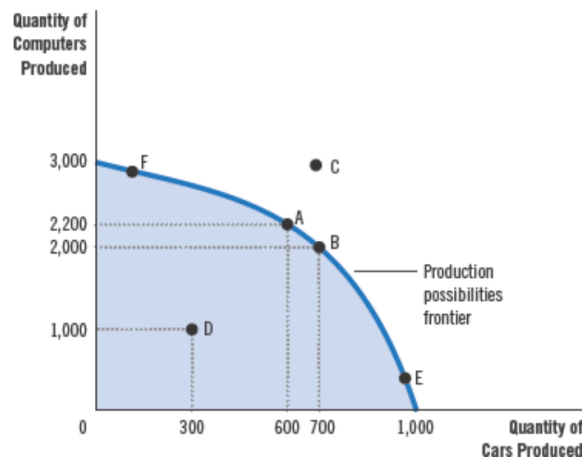
The Circular Flow Diagram

- A visual model of the economy that shows how dollars flow through markets among households and firms
 - Factors of production: labor, land, and capital(buildings, factory, machines, etc)
 - Markets for goods and services: households are buyers and firms are sellers
 - Markets for the factors of production: households are sellers, and firms are buyers.

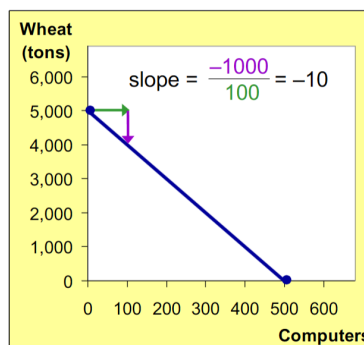


The production possibilities frontier(PPF)

- A graph that shows the combinations of output that the economy can possibly produce with the available factors of production and production technology
 - Efficient outcomes: an outcome if the economy is getting all it can from the scarce resources it has available.
 - * There is no way to produce more of one good without producing less of the other
 - Inefficient outcomes: the economy is producing less than it could from the resources it has available
 - Infeasible outcomes: the economy does not have enough of the factors of production to support that level of output(outside the frontier)
 - Slope of the PPF: the slope measures the **opportunity cost** of x in terms of y



- The shape of the PPF
 - Linear PPF: it indicates that the opportunity cost of producing one good in terms of the other is **constant**.
 - Bowed out(concave to the origin) PPF: it indicates that the opportunity cost of producing one good **increases** as more of it is produced.



- PPF and Economic Growth

- Figure 1: a new pizza assembly line that improves the productive capacity of pizza makers
- Figure 2: when more resources (such as additional workers) are available or improvement in technology for the production of either pizza or wings, the entire PPF shifts upward and outward

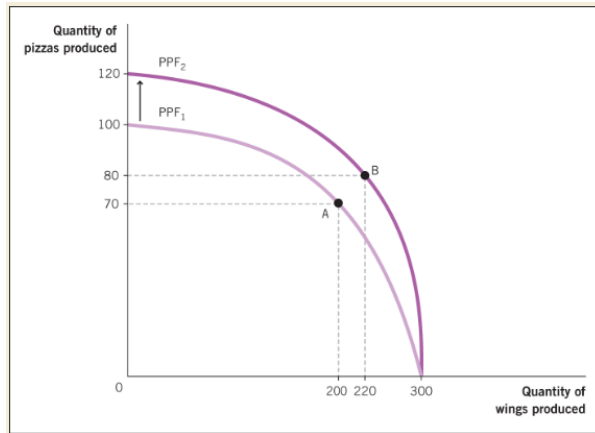


Figure 1

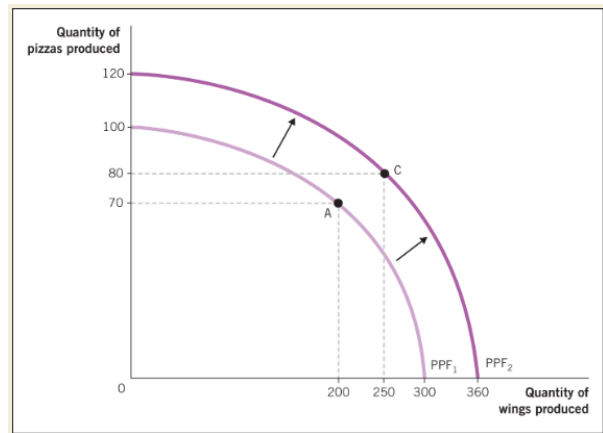


Figure 2

Positive and Normative Analysis

- Positive statements
 - claims that attempt to describe the world as it is (descriptive)
- Normative statements
 - claims that attempt to prescribe how the world should be

Chapter 3. Interdependence & Gain from Trade

- Absolute Advantage
 - the ability to produce a good using fewer inputs than another producer
 - refers to one producer's ability to make more than another produce with the same quantity of resources
- Comparative Advantage
 - the ability to produce a good at a lower opportunity cost than another producer
 - opportunity cost: whatever must be give up to obtain some item
- Trade can benefit everyone because it allows people to specialize in the activities in which they have a comparative advantage
- The price of the trade
 - For both parties to gain from trade, the price at which they trade must lie between their opportunity costs
- e.x) Based on the above table, who has an absolute or comparative advantage in producing pizza and wings?
 - Absolute advantage: Gwen has an absolute advantage in producing both pizzas and wings
 - Comparative advantage: Gwen has a comparative advantage in producing pizza, Blake has a comparative advantage in producing wing

Person	Pizzas	Wings
Gwen	60	120
Blake	24	72

Table 1: Daily production of Pizzas and Wings

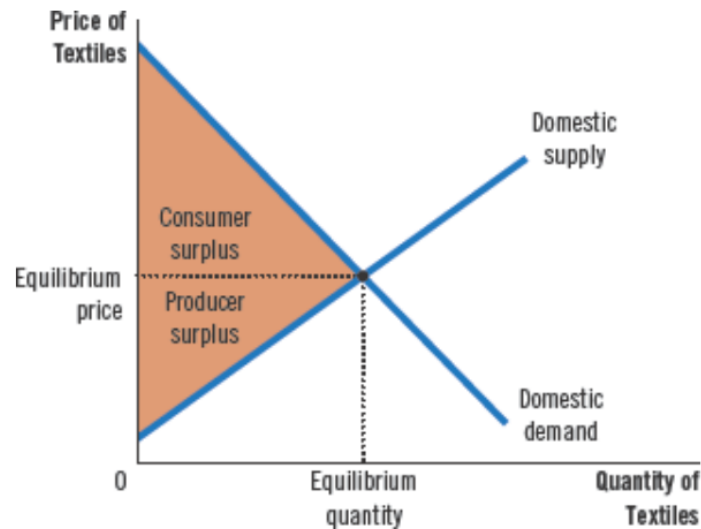
Person	1 Pizza	1 Wings
Gwen	2 wings	1/2 pizza
Blake	3 wings	1/3 pizza

Table 2: Opportunity cost of Pizzas and Wings

Chapter 9. Application to International Trade

The equilibrium without trade

- If there is no international trade, the market for textiles in Isoland consists solely of domestic buyers and sellers (Isoland: no one in this place is allowed to import or export textiles, autarky.)
 - * Autarky: an economic system where a country or region is self-sufficient and limits trade with other countries.



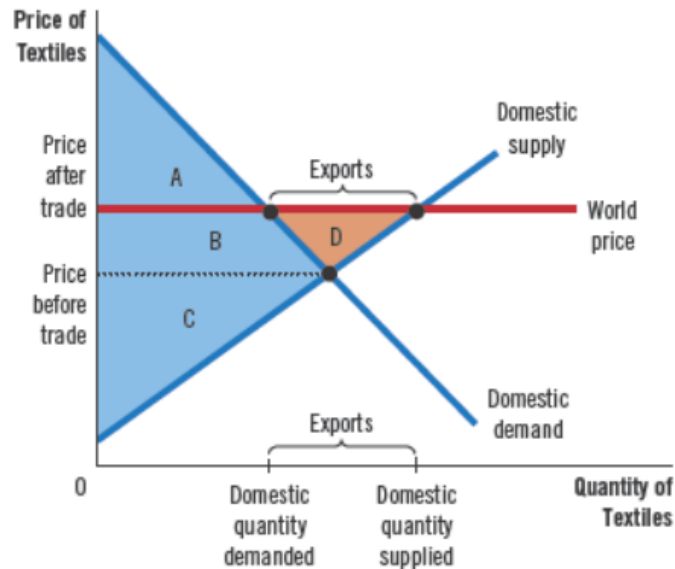
The World Price and Comparative Advantage

- World price: the price of a good that prevails in the world market for that good.
 - * The Isolandians are said to be **price takers** in the world economy. They take the price of goods or services as given by the forces of supply and demand in the world market.
 - * $P_w > P_D$: Isoland will **export** goods or services once trade is permitted
 - * $P_w < P_D$: Isoland will **import** goods or services once trade is permitted

- The gains and losses of an Exporting country
 - * Domestic producers of the good are better off, and domestic consumers of the good are worse off
 - * Trade raises the economic well-being (surplus) of a nation in the sense that the gains of the winners exceed the losses of the losers

	Before Trade	After Trade	Change
Consumer Surplus	$A + B$	A	$-B$
Producer Surplus	C	$B + C + D$	$+(B + D)$
Total Surplus	$A + B + C$	$A + B + C + D$	$+D$

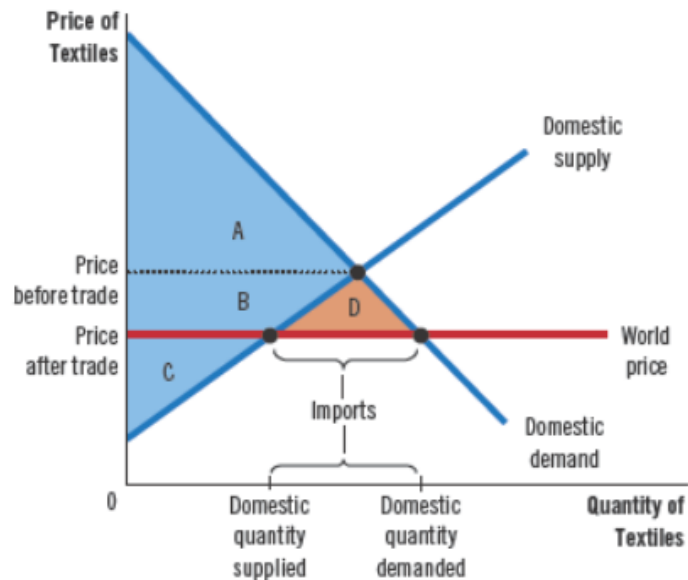
The area D shows the increase in total surplus and represents the gains from trade.



- The gains and losses of an Importing country
 - * Domestic producers of the good are worse off, and domestic consumers of the good are better off
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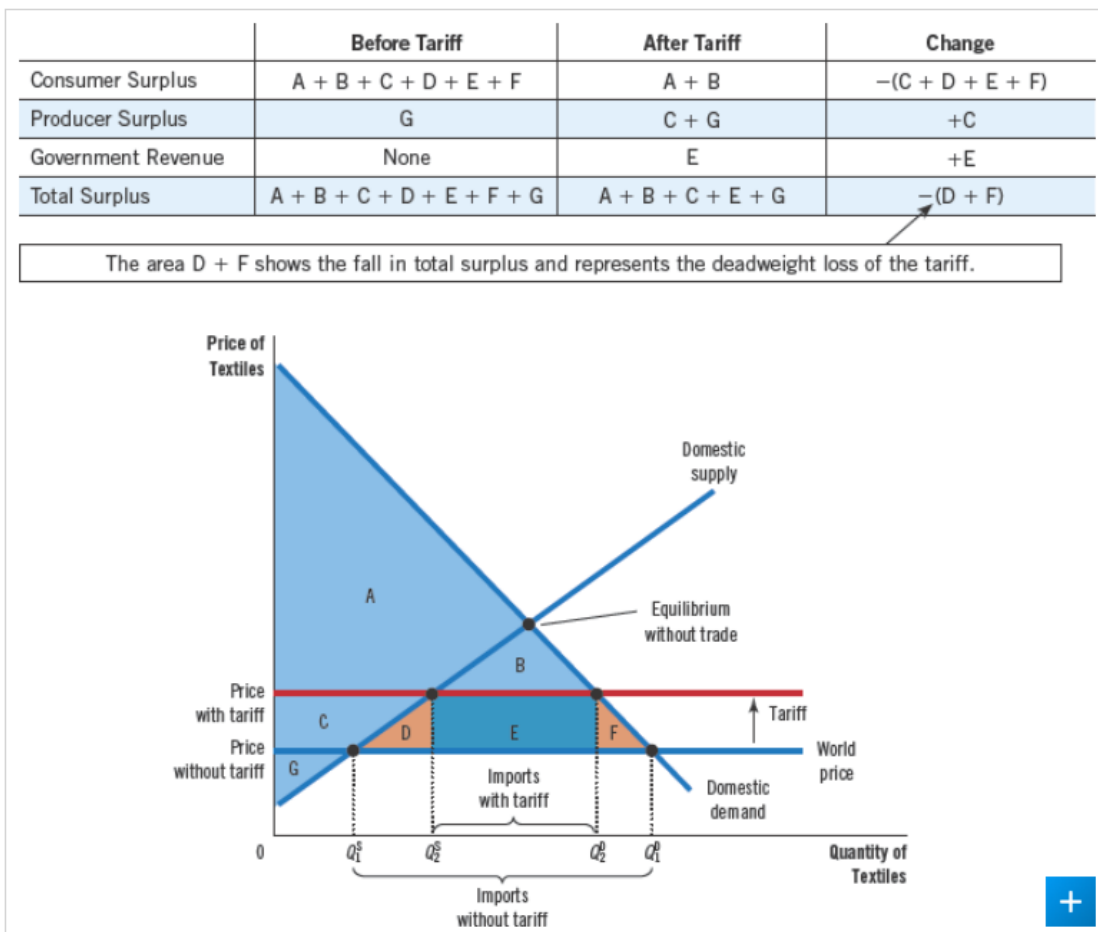
	Before Trade	After Trade	Change
Consumer Surplus	A	A + B + D	+ (B + D)
Producer Surplus	B + C	C	–B
Total Surplus	A + B + C	A + B + C + D	+D

The area D shows the increase in total surplus and represents the gains from trade.



The Effects of a Tariff

- Tariff: a tax on goods produced abroad and sold domestically. A tax on imported goods.
 - * The tariff matters only if Island becomes an importer
 - * A tariff raises the price of imported goods and services above the world price (P_W) by the amount of the tariff.
 - * The change in price (price increase) affects the behavior of domestic buyers and sellers
 - Quantity of demanded: Decrease ($Q_1^D \Rightarrow Q_2^D$)
 - Quantity of supplied: Increase ($Q_1^S \Rightarrow Q_2^S$)
- Deadweight loss: a loss in economic welfare because of the tariff. Tariff distorts incentives and pushes the allocation of scarce resources away from the optimum incentives



Chapter 10. Measuring a Nation's Income

The Measurement of GDP

- Gross Domestic Product(GDP): the market value of all final goods and services produced within a country in a given period
 - (a) “the market value”
 - it uses market prices to translate quantities into a measure of value
 - (b) “of all”
 - it includes all items produced in the economy and sold legally in markets
 - it includes the market value of the housing services that flow from the economy's stock of housing(rental value)
 - it excludes items produced and sold illicitly, such as illegal drugs
 - it excludes most items that never enter the marketplace that are produced and consumed at home
 - (c) “final”
 - GDP includes only the value of final goods because the value of intermediate goods is already included in the prices of the final goods.
 - However, if the intermediate good is added to a firm's inventory, it is taken to be “final” for the moment, and its value as inventory investment is included in GDP
 - (d) “Goods and services”
 - it includes tangible goods and intangible services
 - (e) “produced”
 - it includes goods and services currently produced. It does not include transactions involving items produced in the past
 - ex) the value of the used car is not included in GDP
 - (f) “within a country”
 - GDP measures the value of production within the geographic confines of a country.
 - Items are included in a nation's GDP if they are produced domestically, regardless of the nationality of the producer.
 - (g) “in a given period”
 - GDP measures the value of production that takes place within a specific interval of time
 - Flow variable: is measured over a period of time(GDP, income ...)
 - Stock variable: is measured at a specific point in time(temperature, debt ...)

The Components of GDP

- National Income Account Identity

$$Y = C + I + G + NX \quad (1)$$

- Y : GDP
- C : Consumption
 - * Spending by households on goods and services, with the exception of purchases of new housing
- I : Investment
 - * Investment is the purchase of capital goods that will be used in the future to produce more goods and services (business capital, residential capital, and inventories)
 - * The Purchase of a new house is categorized as an investment.
- G : Government Purchase
 - * Spending on goods and services by local, state, and federal governments.
 - * Transfer payments are not counted as government purchases.
- $NX(X - I)$: Net Exports
 - * Spending on domestically produced goods by foreigners (exports) minus spending on foreign goods by domestic residents (imports)

Real vs. Nominal GDP

- Nominal GDP
 - the production of goods and services valued at current prices
- Real GDP
 - the production of goods and services valued at constant prices
 - Real GDP uses constant base-year prices to value the economy's production
- GDP Deflator
 - a measure of the price level calculated as the ratio of nominal GDP to real GDP times 100

$$GDP \ Deflator = \frac{Nominal \ GDP}{Real \ GDP} \times 100 \quad (2)$$

Year	(P_A)	(P_B)	(Q_A)	(Q_B)
2020 (Base Year)	2.00	1.00	100	200
2021	3.00	1.50	110	220
2022	4.00	2.00	120	250

Table 1: Price and Quantity Data for Apples and Bananas (2020-2022)

Year	Nominal GDP	Real GDP (Base Year 2020)	GDP Deflator
2020	400	400	100
2021	660	440	150
2022	980	490	200

Table 2: Nominal GDP, Real GDP, and GDP Deflator Over Three Years

- GDP is not a perfect measure of well-being
 - Because GDP uses market prices to value goods and services, it excludes the value of almost all activity that takes place outside markets
 - Excludes the quality of the environment, distribution of income, ...
- Gross National Product(GNP)
 - the total market value of all final goods and services produced by the residents of a country within a give time period.
 - It includes the income earned by the country's residents both domestically and internationally, but excludes the income earned by foreign residents within the country's borders

Chapter 11. Measuring the Cost of Living

The Consumer Price Index(CPI)

- CPI
 - a measure of the overall cost of the goods and services bought by a typical consumer
 - (1) Fix the basket
 - (2) Find the prices: find the prices of each of the goods and services in the basket
 - (3) Compute the basket's cost
 - (4) Choose a base year and compute the index

$$CPI = \frac{\text{Cost of basket of goods and services in current year}}{\text{Cost of basket in base year}} \times 100 \quad (3)$$

- (5) Compute the inflation rate

$$\text{Inflation rate in year 2} = \frac{\text{CPI in year 2} - \text{CPI in year 1}}{\text{CPI in year 1}} \times 100 \quad (4)$$

Year	P_H	P_C
2022(Base)	1.00	2.00
2023	2.00	3.00
2024	3.00	4.00

– ex) Basket: 4 hot dogs, 2 cakes

* Cost of basket

- Cost of basket 2022(base) = $\$1 \times 4 + \$2 \times 2 = \$8$
- Cost of basket 2023 = $\$2 \times 4 + \$3 \times 2 = \$14$
- Cost of basket 2024 = $\$3 \times 4 + \$4 \times 2 = \$20$

* CPI

- CPI 2022(base) = $\frac{\$8}{\$8} \times 100 = 100$
- CPI 2023 = $\frac{\$14}{\$8} \times 100 = 175$
- CPI 2024 = $\frac{\$20}{\$8} \times 100 = 250$

* Inflation Rate

- 2023: $\frac{175-100}{100} \times 100 = 75\%$
- 2024: $\frac{250-175}{175} \times 100 = 43\%$

Correcting Economic Variables for the Effects of Inflation

- Dollar Figures from Different Times

$$\text{Amount in today's dollars} = \text{Amount in year T dollars} \times \frac{\text{Price level today}}{\text{Price level in year T}} \quad (5)$$

- Real interest rate

$$\text{Real interest rate}(r) = \text{Nominal interest rate}(i) - \text{Inflation rate}(\pi) \quad (6)$$

Problems with CPI

(1) Substitution Bias

- Over time, some prices rise faster than others. Consumers substitute toward goods that become relatively cheaper, mitigating the effects of price increases

(2) Introduction of New Goods

- The introduction of new goods increases variety, and allows consumers to find products that more closely meet their needs. The CPI misses this effect because it uses a fixed basket of goods

(3) Unmeasured Quality Change

- Quality is hard to measure

Chapter 12. Production and Growth

Productivity: Its Role and Determinants

- Productivity: the quantity of goods and services produced from each unit of labor
 - Real GDP(Y): quantity of output produced
- Determinants of productivity
 - (1) Physical Capital per Worker(K)
 - the stock of equipment and structures that are used to produce goods and services
 - (2) Human capital per Worker(H)
 - the knowledge and skills that workers acquire through education, training, and experience.
 - (3) Natural Resources per Worker(N)
 - the inputs into the production of goods and services that are provided by nature, such as land, rivers, and mineral deposits.
 - (4) Technological Knowledge(A)
 - Society's understanding of the best ways to produce goods and services. It means any advance in knowledge that boosts productivity.

- Production Function

$$Y = A \cdot F(L, K, H, N) \quad (1)$$

- Y: the quantity of output(RGDP)
- L: the quantity of labor
- K: the quantity of physical capital
- H: the quantity of human capital
- N: the quantity of natural resources
- A: the level of technology
- Constant to returns to scale(CRTS)
 - changing all inputs by the same percentage causes output to change by that percentage

for example if all inputs doubled, the outcome also doubled

$$2Y = A \cdot F(2L, 2K, 2H, 2N)$$

- Converting the production function dividing both sides by L(quantity of labor) based on CRTS

$$\frac{Y}{L} = A \cdot F\left(\frac{L}{L}, \frac{K}{L}, \frac{H}{L}, \frac{N}{L}\right)$$

$$\Rightarrow \frac{Y}{L} = A \cdot F\left(1, \frac{K}{L}, \frac{H}{L}, \frac{N}{L}\right)$$

- Y/L: output per worker(real GDP per worker)
- K/L: physical capital per worker
- H/L: human capital per worker
- N/L: natural resources per worker

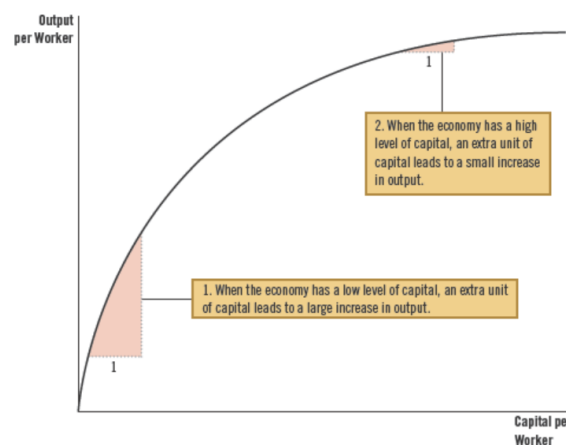
Economic Growth and Public Policy

1. Increase Savings and (therefore) Capital Accumulation

- If the economy produces a large quantity of capital goods today, then tomorrow it will have a larger capital stock and greater capacity to produce goods and services. One way to raise future productivity, then, is to devote more current resources to the production of capital
- However, because resources are scarce, devoting more resources to producing capital requires devoting fewer resources to producing goods and services for current consumption

2. Diminishing Returns and the Catch-Up Effect

- Diminishing Returns(Diminishing marginal product of capital)
 - As the stock of capital rises, the extra output produced from an additional unit of capital falls
 - In other words, when workers already have a large quantity of capital to use in producing goods and services, giving them an additional unit of capital increases productivity only slightly.



- Because of diminishing returns, an increase in the saving rate leads to higher growth only for a while. As the higher saving rate allows more capital to be accumulated, the benefits from additional capital become smaller over time.

- the Catch-up Effect
 - In poor countries: workers have low productivity. In such conditions, small amounts of capital investment can substantially raise productivity.
 - In rich countries: they have high productivity partly because they already have large amounts of capital with which to work. In this case, additional capital investment has a relatively small effect on productivity.

3. Investment from Abroad

- To raise K/L , productivity, wages, and living standards, the government can encourage two types of investment from abroad
 - (a) Foreign direct investment: a capital investment that is owned and operated by a foreign entity(e.g. Toyota Motor Manufacturing, Kentucky)
 - (b) Foreign portfolio investment: an investment financed with foreign money but operated by domestic residents(e.g. the purchase of U.S. Treasury bonds by foreign governments, institutions, and investors. Japan)

4. Education

- A form of investment in human capital(H).
- Positive externality: an externality is the effect of one person's actions on the well-being of a bystander. This can justify the large subsidies to human capital investment through education that we observe in the form of public education
- Brain drain: one problem facing some poor countries is a brain drain - the emigration of highly educated workers to rich countries.

5. Health and Nutrition

- The term human capital can describe another type of investment in people: expenditures that lead to a healthier population - healthier workers are more productive.

6. Property rights and political stability

- Economic prosperity depends in part on favorable political institutions. A country with an efficient court system, honest government officials, and a stable constitution will enjoy a higher standard of living than a country with a poor court system, and corrupt officials.

7. Free trade

- (a) Inward-oriented policies: which aim to increase productivity and living standards by avoiding interaction with the rest of the world(e.g. tariffs)
- (b) Outward-oriented policies: that integrate countries into the world economy and promote trade(e.g. the elimination of restrictions on trade)

8. Research and Development(R&D)

- Knowledge is a public good: once one person discovers an idea, it enters society's pool of knowledge, and other people can freely use it(e.g. patent system, tax incentives for private sector R&D, grant for basic research at universities)

9. Population Growth

- A large population means more workers are available to produce goods and services.

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Chapter 13. Saving, Investment, and the Financial System

Financial Institutions in the U.S. Economy

- Financial system
 - the group of institutions in the economy that help to match one person's savings with another person's investment
 - the financial system moves scarce resources from savers to borrowers.
- Financial markets
 - financial institutions through which savers can *directly* provide funds to borrowers
 - (a) Bond: a certificate of indebtedness. The bond identifies the time at which the loan will be repaid (date of *maturity*) and the *rate of interest* the borrower will pay periodically until the loan matures. Buyers of the bond give their money to a firm in exchange for this promise of interest and eventual repayment of the amount borrowed, called the *principal*
 - (b) Stock: a share of stock represents partial ownership in a firm and is a claim to some of the profits the firm makes.
- Financial Intermediaries
 - financial institutions through which savers can indirectly provide funds to borrowers.
 - the term *intermediary* reflects the role of these institutions in standing between savers and borrowers
 - (a) Banks
 - * A primary job of banks is to take in deposits from people who want to save and use these deposits to make loans to people who want to borrow.
 - * Banks pay depositors interest on their deposits and charge borrowers higher interest on their loans.
 - * Banks help create a special asset, called a medium of exchange, that people can use to engage in transactions (check, debit card, ...)
 - (b) Mutual Funds
 - * an institution that sells shares to the public and uses the proceeds to buy a selection, or portfolio, of various types of stocks, bonds, or both stocks and bonds.
 - * If the value of the portfolio rises, the shareholder benefits; if the value of the portfolio falls, the shareholder suffers the loss.

Saving and Investment in the National Income Accounts

- Recall that gross domestic product(GDP) is both total income in an economy and the total expenditure on the economy's output of goods and services

$$Y = C + I + G + NX \quad (1)$$

- To keep things simple, this chapter assumes that the economy we are examining is **closed economy** ($NX = 0$)

$$Y = C + I + G \quad (2)$$

$$\Rightarrow (Y - C - G) = I \quad (3)$$

$$\Rightarrow S = I \quad (4)$$

– **National Domestic Savings** ($Y - C - G = S$)

- * the total income in the economy that remains after paying for consumption(C) and government purchases(G)
- * By substituting S for $Y - C - G$, we can write the last equation as $S = I$
- * As a results, the national savings(S) equals investment(I)

– **National Domestic Savings = Private Savings + Public Savings**

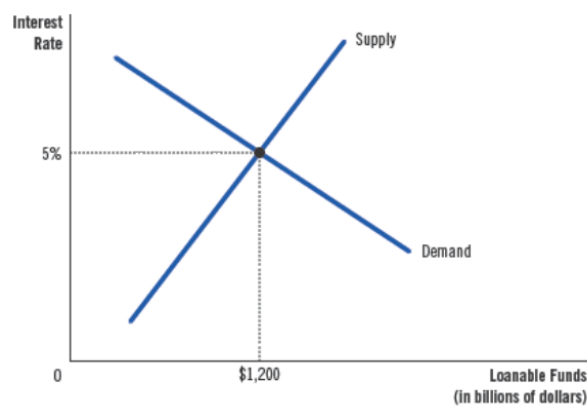
$$S = Y - C - G = I \quad (5)$$

$$S = (Y - T - C) + (T - G) = I \quad (6)$$

- * Private saving($Y - T - C$): the income that households have left after paying for taxes(T) and consumption(C)
 - * Public saving($T - G$): the tax revenue(T) that the government has left after paying for its spending(G)
 - * Budget surplus($T - G > 0$): an excess of tax revenue government spending
 - * Budget deficit($T - G < 0$): a shortfall of tax revenue from government spending
- The equation $S = I$ reveals an important fact: **For the economy as a whole, saving must equal investment**

The Market for Loanable Funds

- Market for loanable funds
 - the market in which those who want to save supply funds and those who want to borrow to invest demand funds
 - the term **loanable funds** refers to the income that people have chosen to save and lend out rather than use for their own consumption and to the amount that investors have chosen to borrow to fund new investment projects.
- Supply and Demand for Loanable Funds
 - (a) **Supply of loanable funds:** comes from people who have an extra income they want to save and lend out
 - (b) **Demand of loanable funds:** comes from households and firms that wish to borrow to make investments.
 - (c) **Interest rate:** the price of a loan. It represents the amount that borrowers pay for loans and the amount that lenders receive on their savings.



- $\Delta I = MPC * \Delta T$
 - MPC(Marginal Propensity to Consume): share of each dollar of disposable income that goes to consumption
 - * e.g.) If $MPC = 0.2(20\%)$, under \$100 income, you use \$20 for consumption and \$80 for saving

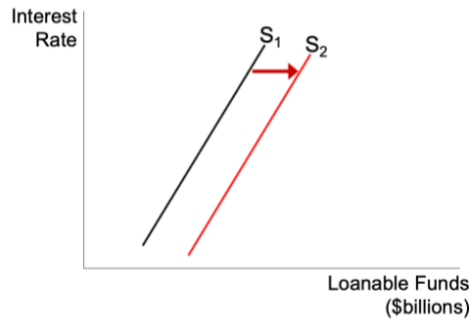
$$\begin{aligned}
 I &= (Y - T - C) + (T - G) \\
 I &= (Y - T) - MPC * (Y - T) + (T - G) \\
 I &= (1 - MPC) * (Y - T) + (T - G), \quad S_p = (1 - MPC)(Y - T) \quad (7)
 \end{aligned}$$

$$\begin{aligned}
 \Delta I &= (1 - MPC) * (\Delta Y - \Delta T) + (\Delta T - \Delta G) \\
 \Delta I &= (1 - MPC) * (-\Delta T) + (\Delta T), \quad \Delta Y = \Delta G = 0 \\
 \Delta I &= MPC * \Delta T \quad (8)
 \end{aligned}$$

- Shift the loanable funds supply right

(1) An increase in private savings: $Y - T - C = (1 - MPC)(Y - T)$

- Decrease in MPC($\downarrow MPC$)
- Lower taxes($\downarrow T$)
- Larger income($\uparrow Y$)



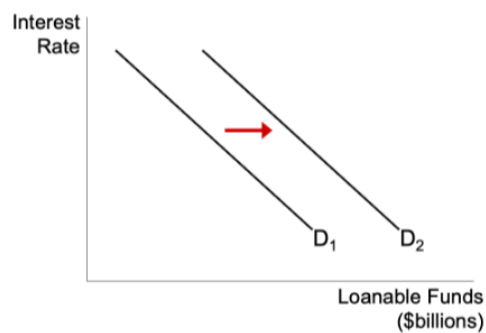
(2) An increase in public savings: $T - G$

- Increase in taxes($\uparrow T$)
- Decrease in government expenditure($\downarrow G$)
- Everything is equal, an increase in taxes increases national savings by $\Delta I = \Delta S = MPC * \Delta T$

- Shift the loanable funds demand right

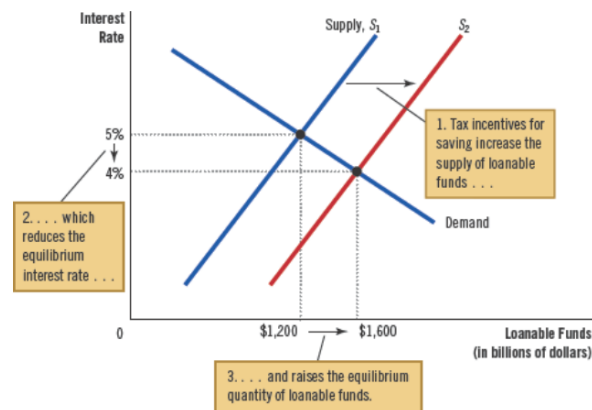
(1) An increase in investment

- Better investment opportunities
- Investment tax credits
- An increase in firms' optimism

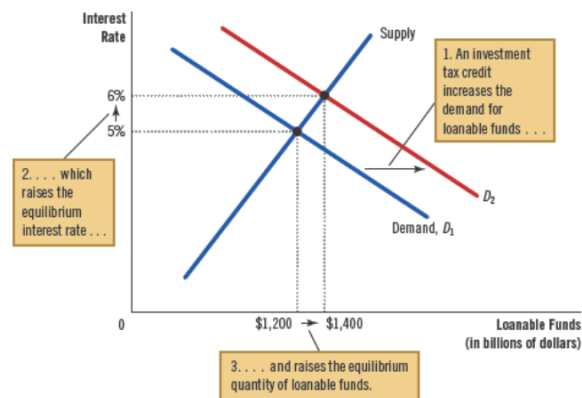


Government policies

1. Saving Incentives

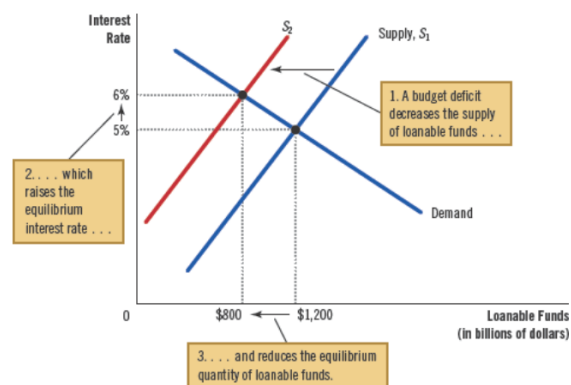


2. Investment Incentives



3. Government Budget Deficit and Surpluses

- The Crowding-Out Effect: increase in budget deficit causes fall in investment. The government borrows to finance its deficit, leaving less funds available for investment.



Chapter 15. Unemployment

Identifying Unemployment

- The Bureau of Labor Statistic(BLS) places each adult(**age 16 and older**) in surveyed households into one of three categories

(1) Employed

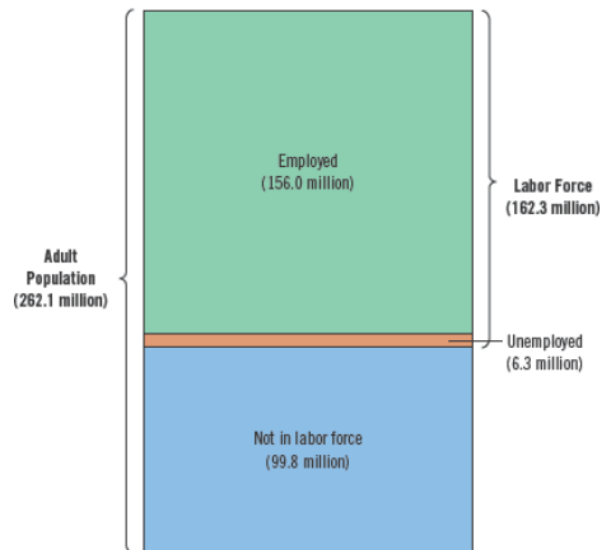
- Include those who worked as paid employees, worked in their own business, or worked without pay in a family member's business.
- Both full-time and part-time workers are counted.
- Those who had jobs from which they were temporarily absent because of vacation, illness, bad, weather, or similar reasons also included in this category

(2) Unemployed

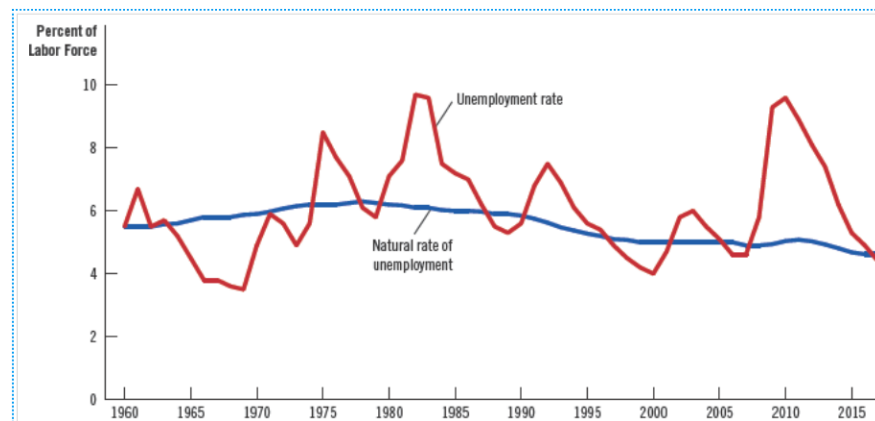
- Include those who were not employed, were available work, and had tried to find employment during the previous **four weeks** are in this group
- Also includes those waiting to be recalled to a job after a **temporary layoff**.

(3) Not in the labor force

- Includes those who fit into neither of the unemployed and employed categories, such as full-time students, homemakers, and retirees.



- Unemployment rate and Labor-force participation rate
 - Labor force = # of employed + # of unemployed
 - * the total number of workers, including both the employed and unemployed
 - Unemployment rate = $\frac{\# \text{ of unemployed}}{\text{Labor force}} \times 100$
 - * the percentage of the labor force that is unemployed
 - Labor-force participation rate = $\frac{\text{Labor force}}{\text{Adult population}} \times 100$
 - * the percentage of the adult population that is in the labor force
 - e.g. 156 million people were employed, and 6.3 million people were unemployed. What's the number of the labor force? Unemployment rate? Labor-force participation rate if the adult population is 261 million?
- Natural rate of unemployment and Cyclical unemployment



- Natural rate of unemployment: the normal rate of unemployment around which the unemployment rate fluctuates
 - Cyclical unemployment: the deviation of unemployment from its natural rate
 - * Unemployment caused by economic downturns and business cycle fluctuations (recessions)
- Does the unemployment rate measure what we want it to measure?
 - Among those who are not working, it is hard to tell who is unemployed and who is not in the labor force.
 - Unemployment rate excludes discouraged workers
 - * Discouraged workers: individuals who would like to work but have given up looking for a job
 - Unemployment rate does not distinguish between full-time and part-time work
 - Some people misreport their work status in the BLS survey.

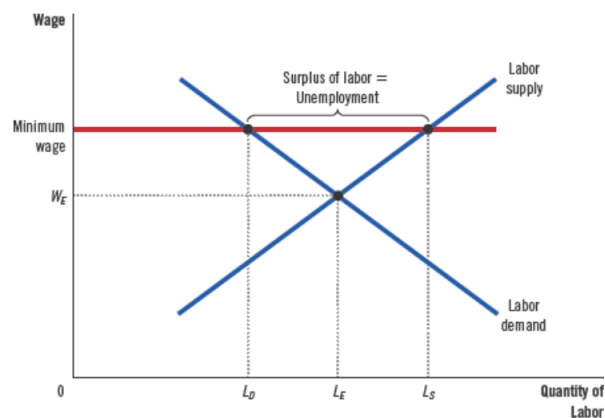
- In judging how serious the problem of unemployment is, one question to consider is whether unemployment is typically a short-term or long-term condition.
 - If unemployment is short-term, one might conclude that it is not a big problem.
 - Yet if unemployment is long-term, one might conclude that it is a serious problem.
 - Most people who become unemployed find a job quickly, but at any given moment, most of the unemployed people you see have been out of work for a long time.
 - * e.g. If you visited an unemployment office every week for a year to survey the unemployed. Each week you find four unemployed workers. Three are the same individuals for the whole year, while the fourth person changes every week. Based on this experience, would you say that unemployment is typically short-term or long-term?
 - 95 percent of unemployment end in one week
 - 75 percent of the unemployment at any moment is attributable to those who are unemployed for a full year.
- Why are there always some people unemployed?
 - Unemployment never falls to zero but, instead, fluctuates around its natural rate
 - **Frictional unemployment:** unemployment that results because it takes time for workers to search for the jobs that best suit their tastes and skills. It is often thought to explain relatively short spells of unemployment
 - (1) Job search
 - **Structural unemployment:** Unemployment that results because the number of jobs available in some labor markets is insufficient to provide a job for everyone who wants one. Usually long-term.
 - (2) Minimum-Wage Laws
 - (3) Unions and Collective Bargaining
 - (4) The theory of Efficiency Wages

(1) Job Search

- The process by which workers find appropriate jobs given their tastes and skills
- Changes in the composition of demand among industries or regions are called **sectoral shifts**. Because it takes time for workers to search for jobs in the new sectors, sectoral shifts cause unemployment
- Public policy may be able to reduce the time it takes unemployment workers to find new jobs, thereby lowering the economy's natural rate of unemployment
 - Government-run employment agencies: which give out information about job vacancies.
 - Training programs: which aim to ease workers' transition from declining to growing industries and help disadvantaged groups escape poverty
- **Unemployment Insurance:** a government program that partially protects workers' incomes when they become unemployed. It helps the unemployed to find better job matches. **However, it can increase the amount of frictional unemployment, without intending to do so.**

(2) Minimum-Wage Laws

- When a minimum wage is binding, it forces the wage to remain above the level that balances supply and demand.
- Minimum-wage laws matter most for those with low levels of skill and experience, such as teenagers
- If the wage is kept above the equilibrium level for any reason, unemployment results



(3) Union

- A worker association that bargains with employers over wages, benefits, and working conditions
- **Collective bargaining:** the process by which unions and firms agree on the terms of employment. In these cases, wages are not determined by the equilibrium of supply and demand in competitive labor markets.
- When unions raise the wage above equilibrium, the quantity of labor demanded falls, and unemployment results.

(4) Efficiency Wages(The Theory of Efficiency Wages)

- Above-equilibrium wages paid by firms to increase worker productivity. Four reasons why firms might pay efficiency wages
 - (a) Worker health: a firm may find it profitable to pay high wages to ensure its workers are healthy and productive
 - (b) Worker turnover: a firm can reduce turnover by paying higher wages
 - (c) Worker quality: when a firm pays high wages, it attracts a better pool of workers to apply for its jobs and thereby increases the quality of its workforce
 - (d) Worker effort: firms may raise wages above the equilibrium level to incentivize workers not to shirk their responsibilities

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Chapter 16. The Monetary System

The Meaning of Money

- Money
 - the set of assets in an economy that people regularly use to buy goods and services
- The Functions of Money
 - (1) A medium of exchange: an item that buyers give to sellers when they want to purchase goods and services
 - (2) Unit of account: the yardstick people use to post prices and record debts
 - (3) Store of value: an item that people can use to transfer purchasing power from the present to the future
- Liquidity: the ease with which an asset can be converted into the economy's medium of exchange. Money is the economy's medium of exchange, it is the most liquid asset available.
- The kinds of Money
 - Commodity money: money that takes the form of a commodity with intrinsic value(e.g. gold, silver, salt)
 - Fiat Money: money without intrinsic value that is used as money by government decree
- Money in the U.S. Economy
 - M1: includes currency, demand deposit at banks, traveler's checks, and other checkable deposits
 - M2: M1 + savings deposits, small time deposits, money market mutual funds,

The Federal Reserve System

- Federal Reserve
 - the central bank of the United States
 - central bank is an institution designed to oversee the banking system and regulate the quantity of money in the economy

- Roles of Fed
 - Regulate banks and ensure the health of the banking system
 - * Monitors each bank's financial condition
 - * The Fed makes loans to banks when banks themselves need funds.
 - Monetary policy
 - * the setting of the money supply by policymakers in the central bank
 - * The Fed promotes economic prosperity by controlling the quantity of money available in the economy in the short run and it's connected to the level of interest rate(FOMC)

The Maturity Mismatch

- Borrow short and lend long
 - Borrow short: Commercial banks borrow money from depositors who can withdraw them without notice at any ATM
 - Lend long: They lend money to households or firms in the form of long-term loans (mortgages)
- Bank Run: if too many depositors withdraw money at the same time, bank is not able to repay all depositors and people run to banks to be the first in the line(eg. 2023, Silicon Valley Bank (SVB))

Fractional Reserve Banking System

- Reserves
 - deposits that banks have received but have not loaned out
 - if banks hold all deposits in reserve, banks do not influence the supply of money

First National Bank			
Assets		Liabilities	
Reserves	\$100.00	Deposits	\$100.00

- Fractional-reserve banking
 - a banking system in which banks hold only a fraction of deposits as reserves
 - Reserve ratio(R): the fraction of deposits that banks hold as reserves
 - Reserve requirement: the Fed traditionally sets a minimum amount of reserves that banks must hold
 - Excess reserves: reserves that are above the legal minimum

- The Money Multiplier(= the reciprocal of the reserve ratio($\frac{1}{R}$))
 - the amount of money that results from each dollar of reserves
 - The higher the reserve ratio, the less of each deposit banks loan out $\Rightarrow$, and the smaller the money multiplier.
 - ex) When the reserve ratio is 10%, money multiplier is $\frac{1}{10\%} = \frac{1}{0.1} = 10$. In this case, the \$100 of reserves generates \$1,000 of money.

First National Bank			
Assets		Liabilities	
Reserves	\$10.00	Deposits	\$100.00
Loans	90.00		

Second National Bank			
Assets		Liabilities	
Reserves	\$ 9.00	Deposits	\$90.00
Loans	81.00		

Third National Bank			
Assets		Liabilities	
Reserves	\$ 8.10	Deposits	\$81.00
Loans	72.90		

- Bank Capital, Leverage and Financial Crisis
 - Bank Capital: the resources a bank's owners have put into the institution. In addition to making loans and holding reserves, it can buy financial securities, such as stocks and bonds
 - Leverage: the use of borrowed money to supplement existing funds for investment purposes
 - Leverage Ratio: the ratio of assets to bank capital
 - Capital Requirement: a government regulation specifying a minimum amount of bank capital

More Realistic National Bank			
Assets		Liabilities and Owners' Equity	
Reserves	\$200	Deposits	\$800
Loans	700	Debt	150
Securities	100	Capital (owners' equity)	50

- Leverage case (1)
 - If bank's asset increase by 5%(\$1,000 $\Rightarrow$ \$1,050), then the bank capital increase from \$50 to \$100.
 - $\Rightarrow$ When the leverage ratio is 20. A 5-percent appreciation in asset values **increases** the owner's equity by $100\%(\frac{\$100-\$50}{\$50} \times 100)$
- Leverage case (2)
 - If bank's asset decrease by 5%(\$1,000 $\Rightarrow$ \$950), then the bank capital decreases from \$50 to \$0.
 - $\Rightarrow$ When the leverage ratio is 20. A 5-percent fall in asset values **reduces** the owner's equity by $100\%(\frac{\$50-\$0}{\$50} \times 100)$
 - $\Rightarrow$ If the value of assets were to fall by more than 5%, the bank's assets would fall below its liabilities, becoming **insolvent**.

The Fed's Tools of Monetary Control

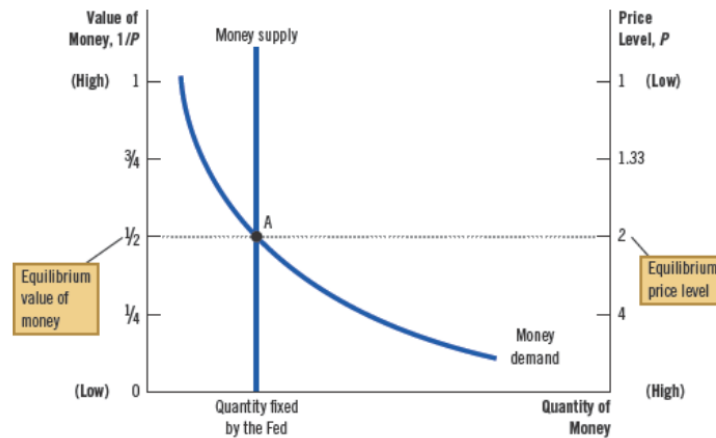
- The Federal Open Market Committee(FOMC)
 - Open-market operation: the purchase and sale of U.S. government bonds
 - * Increase the money supply: the Fed can create dollars and use them to buy government bonds from the public in the bond markets $\Rightarrow$ the dollars go into the public's hands
 - * Decrease the money supply: the Fed can sell government bonds from its portfolio to the public in the bond markets $\Rightarrow$ the dollars the Fed receives for the bonds are out of public hands
- Fed lending to banks
 - Banks borrow from the Fed when they do not have enough reserves on hand either to satisfy bank regulators, meet depositor withdrawals, make new loans, or for some other business reason.
 - Discount rate: the interest rate on the loans that the Fed makes to banks
 - * A higher discount rate: discourages banks from borrowing from the Fed, decreasing the quantity of reserves in the banking system and the money supply.
 - * A lower discount rate: encourages banks to borrow from the Fed, increasing the quantity of reserves and the money supply
- Reserve Requirements
 - regulations on the minimum amount of reserves that banks must hold against deposit
 - * Increase in reserve requirements: raises the reserve ratio, lowers the money multiplier, and reduces the money supply.
 - * Decrease in reserve requirements: lowers the reserve ratio, raises the money multiplier, and increases the money supply
- Interest rates on reserves
 - the interest rate paid to banks on the reserves held in deposit at the Fed

- Federal Funds rate
 - the interest rate at which banks make overnight loans to one another
 - The Fed has set a target for the federal funds rate.
 - * A decrease in the target for the federal funds rate: an expansion in the money supply
 - * An increase in the target for the federal funds rate: a contraction in the money supply

Chapter 17. Money Growth and Inflation

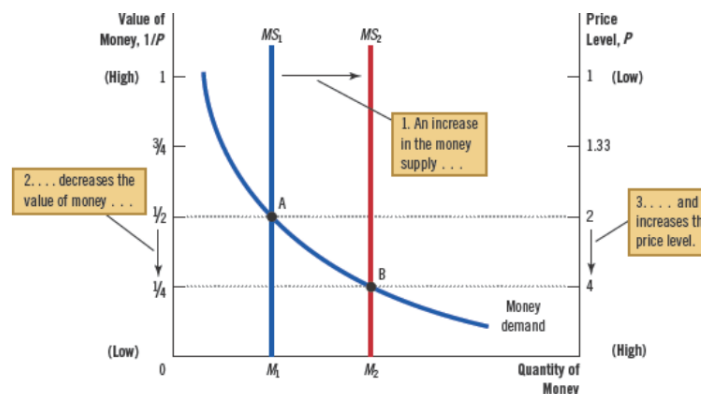
Money Supply, Money Demand, and Monetary Equilibrium

- The level of prices and the value of money
 - Mathematical expression
 - * P : the price of goods and services measured in terms of money
 - * $\frac{1}{P}$: the value of money measured in terms of goods and services
 - * ex) If \$100 is the price of a basket of goods and services, then $\frac{1}{100}$ is the quantity of goods and services that can be bought with \$1
- What factors influence money supply and money demand?
 - Money Supply
 - * The quantity of money supplied is a policy variable that the Fed controls
 - Money Demand
 - * The demand for money reflects how much wealth people want to hold in liquid form.
 - * The quantity of money demanded depends on the interest rate that a person could earn by using the money to buy an interest-bearing bond rather than leaving it in his wallet or low-interest checking account.
 - * How much money do they choose to hold for buying the goods and services depends on the prices of those goods and services
 - ⇒ A higher price level(a lower value of money) increases the quantity of money demanded
- In the **long-run**, money supply and money demand are brought into equilibrium by the overall level of prices
 - If the price level is above the equilibrium level ⇒ people will want to hold more money than the Fed has created ⇒ the price level must fall to balance supply and demand
 - If the price level is below the equilibrium level ⇒ people will want to hold less money than the Fed has created ⇒ the price level must rise to balance supply and demand



- The effects of a Monetary Injection

- What happens if the Fed alters the money supply using the tools discussed in the previous chapter?
 - * Monetary injection: shift the supply curve to the right $MS_1 \Rightarrow MS_2$
 - * The equilibrium moves from point A to B
 - * The value of money decreases, and the equilibrium price level increases
 $\Rightarrow$ When an increase in the MS makes dollars more plentiful, the result is an increase in the price level and a decrease in the value of each dollar.



- Quantity theory of money

- A theory asserting that the quantity of money available determines the price level and that the growth rate in the quantity of money available determines the inflation rate

- The classical dichotomy and monetary neutrality
 - Classical dichotomy: the theoretical separation of nominal variables and real variables
 - * Nominal variables: variables measured in monetary units(ex. nominal GDP, Nominal wage)
 - * Real variables: variables measured in physical units(ex. relative price, real wage, real GDP)
 - Monetary neutrality: the proposition that changes in the money supply do not affect real variables
 - ⇒ ex) If the money supply gets doubled, the price level doubles, the dollar wage doubles, and all other dollar values double. Then real variables, such as production, employment, real wages, and real interest rates.
 - ⇒ **In the long run**, the money supply affects nominal variables but not affect real variables.

- Velocity and the Quantity Equation

- Velocity of money(V): the rate at which money changes hands

$$V = (P \times Y)/M$$

$$M \times V = P \times Y$$

- * P: price level
 - * Y: real GDP
 - * M: the quantity of money
 - The velocity of money(V) is relatively stable over time. And the money supply(M) is neutral so that it does not affect Y
 - ⇒ Thus, P changes by same percentage as M

- Inflation Tax

- When tax revenue is inadequate and the ability to borrow is limited ⇒ government may print money to pay for its spending
 - When the government prints money, the money supply increases, the price level increases, and the dollars in your wallet become less valuable ⇒ This inflation is like a tax on everyone who holds money

- The Fisher Effect

- Fisher equation

$$\text{Real interest rate}(r) = \text{Nominal interest rate}(i) - \text{inflation rate}(\pi)$$

$$\text{Real interest rate}(r) + \text{inflation rate}(\pi) = \text{Nominal interest rate}(i)$$

- When a central bank increases the rate of money growth, the long-run result is an equal increase in the inflation rate and the nominal interest rate. It's based on the concept of money neutrality in the long run

- Shoeleather cost
 - the resources wasted when inflation encourages people to reduce their money holdings
 - the time and convenience you must sacrifice to keep less money on hand than you would if there were no inflation
- Menu cost
 - the costs of changing prices
 - inflation increases the menu costs that firms must bear because it includes the cost of deciding on new prices, printing new price lists and catalogs, sending these new price lists and catalogs to dealers and customers ...

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Chapter 18. Open-Economy Macroeconomics: Basic Concepts

- Closed Economy($Y = C + I + G$)
 - an economy that does not interact with other economies in the world
- Open Economy($Y = C + I + G + NX$)
 - an economy that interacts freely with other economies around the world
 - An open economy interacts with other economies in two ways
 - (1) Buy and sell goods and services in world product markets
 - (2) Buy and sell capital assets such as stocks and bonds in world financial markets

The Flow of Goods: Exports, Imports, and Net Exports

- Exports(X): goods and services that are produced domestically and sold abroad
- Imports(M): goods and services that are produced and sold domestically
- Net Exports($NX = X - M$): spending on domestically produced goods by foreigners - spending on foreign goods by domestic residents
 - Trade Balance: the value of a nation's exports - the value of its imports; also called net exports
 - Trade Surplus($NX > 0$): an excess of exports over imports
 - Trade Deficit($NX < 0$): an excess of imports over exports
 - Balanced Trade($NX = 0$): a situation in which $X = M$
- Factors
 - Consumer tastes for domestic and foreign goods
 - The prices of goods at home and abroad
 - The exchange rates at which people can use domestic currency to buy foreign currencies
 - The incomes of consumers at home and abroad
 - The cost of transporting goods from country to country
 - Government policies toward international trade

The Flow of Financial Resources: Net Capital Outflow(NCO)

- Net Capital Outflow(NCO)
 - the purchase of foreign assets by domestic residents - the purchase of domestic assets by foreigners
 - The flow of capital takes two forms
 - (1) Foreign direct investment(FDI)
 - (2) Foreign portfolio investment
 - Factors
 - * The real interest rates paid on foreign assets
 - * The real interest rates paid on domestic assets
 - * The perceived economic and political risks of holding assets abroad
 - * The government policies that affect foreign ownership of domestic assets

The Equality of Net Exports and Net Capital Outflow

$$Net\ Capital\ Outflow(NCO) = Net\ Exports(NX)$$

- If $NX > 0$, it is selling more goods and services to foreigners. Then, it must be used to buy foreign assets. Capital is flowing out of the country($NCO > 0$)
- If $NX < 0$, it is buying more goods and services from foreigners. Then, it must be selling assets abroad. Capital is flowing into the country($NCO < 0$)
- e.g.) MS sells software to Japan at ¥10,000

Saving, Investment, and their relationship to International Flows

In an open economy :

$$Y = C + I + G + NX$$

$$\Rightarrow Y - C - G = I + NX$$

$$\Rightarrow S(National\ Saving) = I + NX$$

$$\Rightarrow S = I + NCO$$

- $S = I + NCO$
 - A nation's saving must equal its domestic investment(I) plus its NCO
 - When a U.S. household saves a dollar of its income for the future, that dollar can be used to finance either the accumulation of domestic capital or the purchase of foreign capital
 - $S > I$: NCO is positive, indicating that the nation is using some of its saving to buy assets abroad
 - $S < I$: NCO is negative, indicating that foreigners are financing some of this investment by purchasing domestic assets

The Prices for International Transactions: Real and Nominal Exchange Rates

- Nominal Exchange Rates

- the rate at which a person can trade the currency of one country for the currency of another
 - * e.g.) 1 EUR = 1.08 USD, 1 USD = 1,470 WON
- Appreciation: an increase in the value of a currency as measured by the amount of foreign currency it can buy
- Depreciation: a decrease in the value of a currency as measured by the amount of foreign currency it can buy
- e.g.) 1 EUR = 1.00 USD $\Rightarrow$ 1 EUR = 1.08 USD
 - * Euro appreciated, and USD depreciated
- e.g.) 1 USD = 1,000 WON $\Rightarrow$ 1 USD = 1,470 WON
 - * USD appreciated, and WON depreciated

- Real Exchange Rates

- the rate at which a person can trade the goods and services of one country for the goods and services of another

$$\text{Real Exchange Rate} = \frac{\text{Nominal Exchange Rate} \times \text{Domestic Price}}{\text{Foreign Price}} = \frac{e \times P}{P^*}$$

- * If nominal exchange rate \$1 = €1, coffee in U.S. is \$3 and coffee in Europe is €4. Then what's the real exchange rate?
- * What happens if the nominal exchange rate becomes \$1 = €2?
- * What happens if the nominal exchange rate becomes \$1 = €0.5?
- Real exchange rate < 1: cheaper in U.S.
- Real exchange rate > 1: cheaper in a foreign country
- Real exchange rate = 1: equal price(purchasing power parity)

A First Theory of Exchange-Rate Determination: Purchasing-Power Parity

- Law of one price

- the notion that a good should sell for the same price in all markets

- Arbitrage

- The process of taking advantage of price differences for the same item in different markets is called arbitrage

- Purchasing-Power Parity(PPP)

$$\text{Real exchange rate} = 1 = \frac{e \times P}{P^*}$$

$$\Rightarrow P^* = e \times P$$

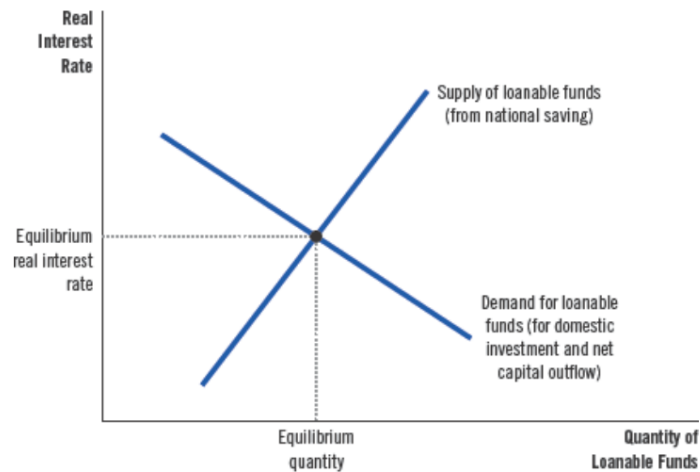
$$\Rightarrow e = \frac{P^*}{P}$$

- a theory of exchange rates that says a unit of any given currency should be able to buy the same quantity of goods in all countries
 - according to the theory of purchasing-power parity, the nominal exchange rate(e) must reflect the price levels in those countries
- Limitation of the PPP
 - First, many goods are not easily traded. So the price differences on such goods cannot be arbitrated away
 - Second, domestic goods are not perfect substitutes. Consumers have different preferences and it can change over time

Chapter 19. A Macroeconomic Theory of the Open Economy

Supply and Demand for Loanable Funds and for Foreign-Currency Exchange

- The Market for Loanable Funds
 - Under the open economy, the simplified model assumes that the financial system has only one market: the market for loanable funds.
 - Identity equation: $S = I + NCO$
 - * When a nation saves a dollar of its income(S), it can use that dollar to finance the purchase of domestic capital(I) or to finance the purchase of an asset abroad(NCO)
 - * The supply of loanable funds: national saving(S)
 - * The demand for loanable funds: domestic investment(I) and net capital outflow(NCO)

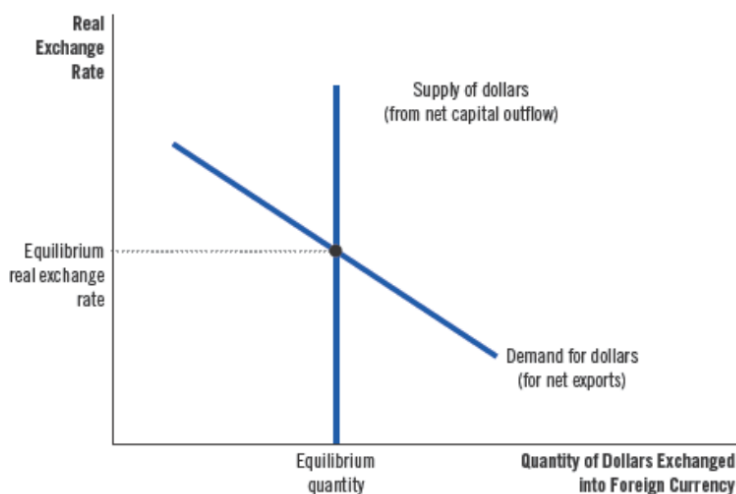


- A country's real interest rate(r) affects S , I and NCO
 - * In an open economy, the demand for loanable funds comes not only from those who want loanable funds to buy domestic capital goods but also from those who want loanable funds to buy foreign assets
 - * At the equilibrium interest rate(r^*), the amount that people want to save(Q_s) balances the desired quantities of domestic investment and net capital outflows(Q_d)

- The Market for Foreign-Currency Exchange

- Identity equation: $NCO = NX$

- * the imbalance between the purchase and sale of capital assets abroad (NCO) equals the imbalance between exports and imports of goods and services (NX)
- * if $NX > 0$: foreigners are buying more U.S. goods and services than Americans are buying foreign goods and services $\Rightarrow$ With the foreign currency, they buy foreign assets, so U.S. capital is flowing abroad ($NCO > 0$)
- * if $NX < 0$: Americans are spending more on foreign goods and services than they are earning from selling goods and services abroad $\Rightarrow$ Some of this spending must be financed by selling American assets abroad ($NCO < 0$)



- * NCO : the quantity of dollars supplied for the purpose of buying foreign assets
- * NX : the quantity of dollars demanded for the purpose of buying U.S. net exports of goods and services
- The real exchange rate (RER) balances the supply and demand in the market for foreign-currency exchange
 - * For example, a higher RER makes U.S. goods more expensive $\Rightarrow$ reduces the quantity of dollars demanded to buy those goods
- The supply curve is **vertical**
 - * this is because the quantity of dollars supplied for NCO does not depend on the RER
 - * NCO depends on the real interest rate (r)
 - * Changes in the exchange rate influence not only the cost of buying foreign assets but also the benefit of owning them, and these two effects offset each other

Equilibrium in the Open Economy

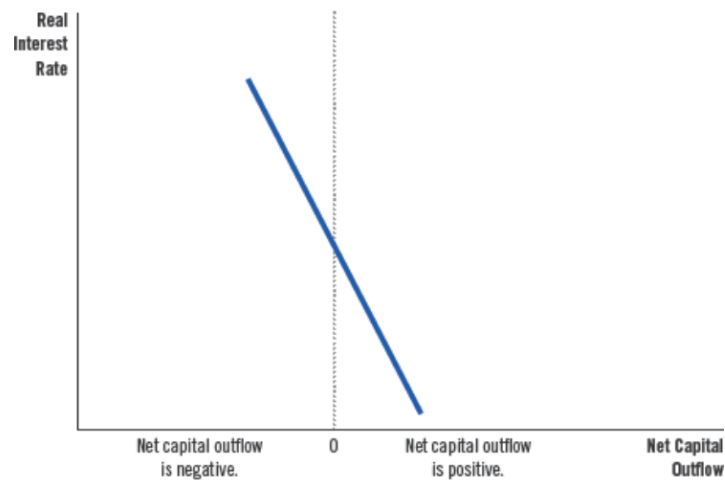
- Net Capital Outflow: The Link between the Two Markets

(1) Market for Loanable funds: $S = I + NCO$

(2) Market for Foreign-Currency Exchange: $NCO = NX$

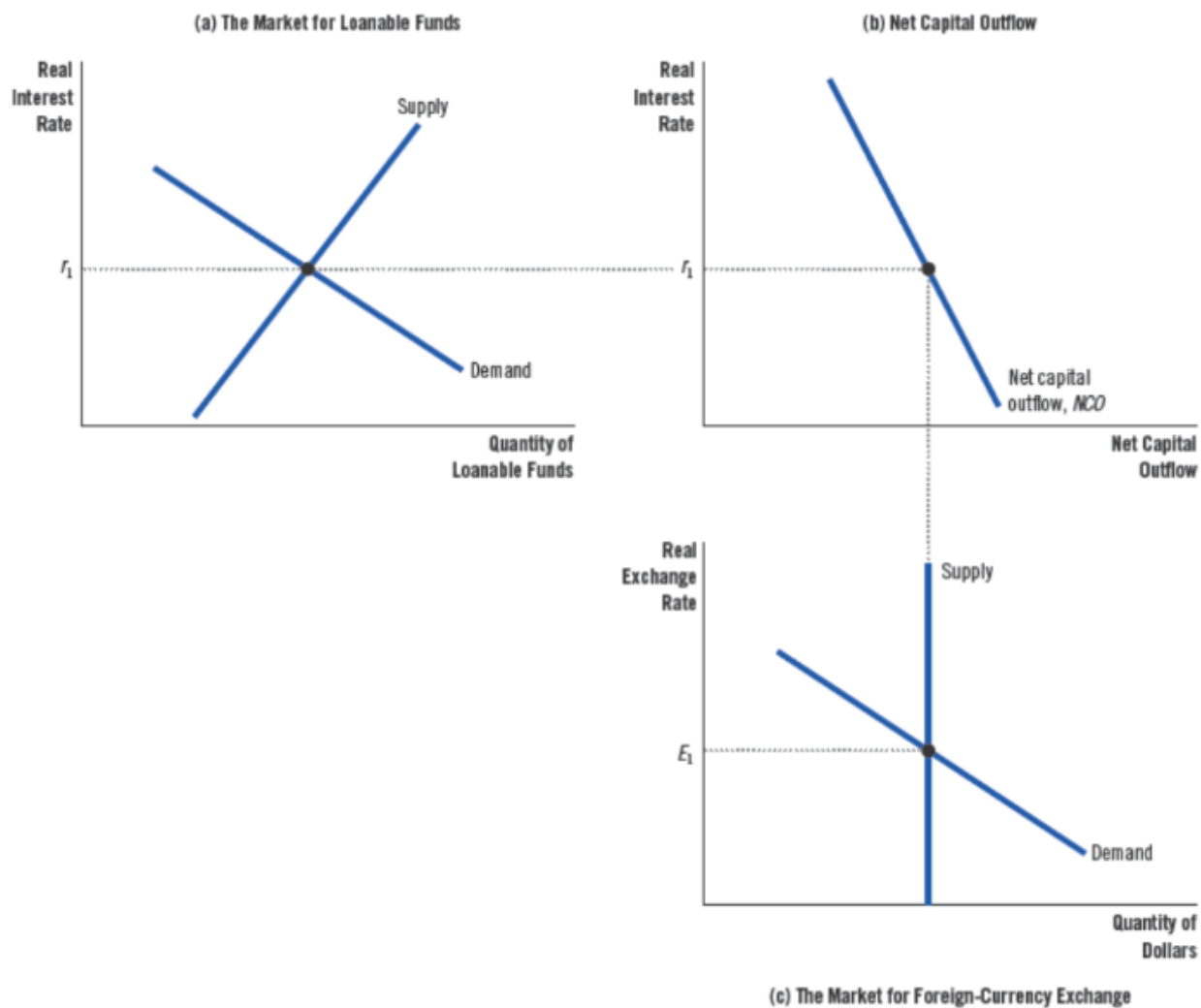
- (1) An American who wants to buy an asset abroad must finance this purchase by obtaining resources in the U.S. market for loanable funds
- (2) An American who wants to buy an asset in another country must supply dollars to exchange them for the currency of that country

- The key determinant of NCO is the real interest rate(r)



- * When $r \uparrow$: domestic assets more attractive $\Rightarrow$ reduces NCO
- * When $r \downarrow$: domestic assets less attractive $\Rightarrow$ increases NCO

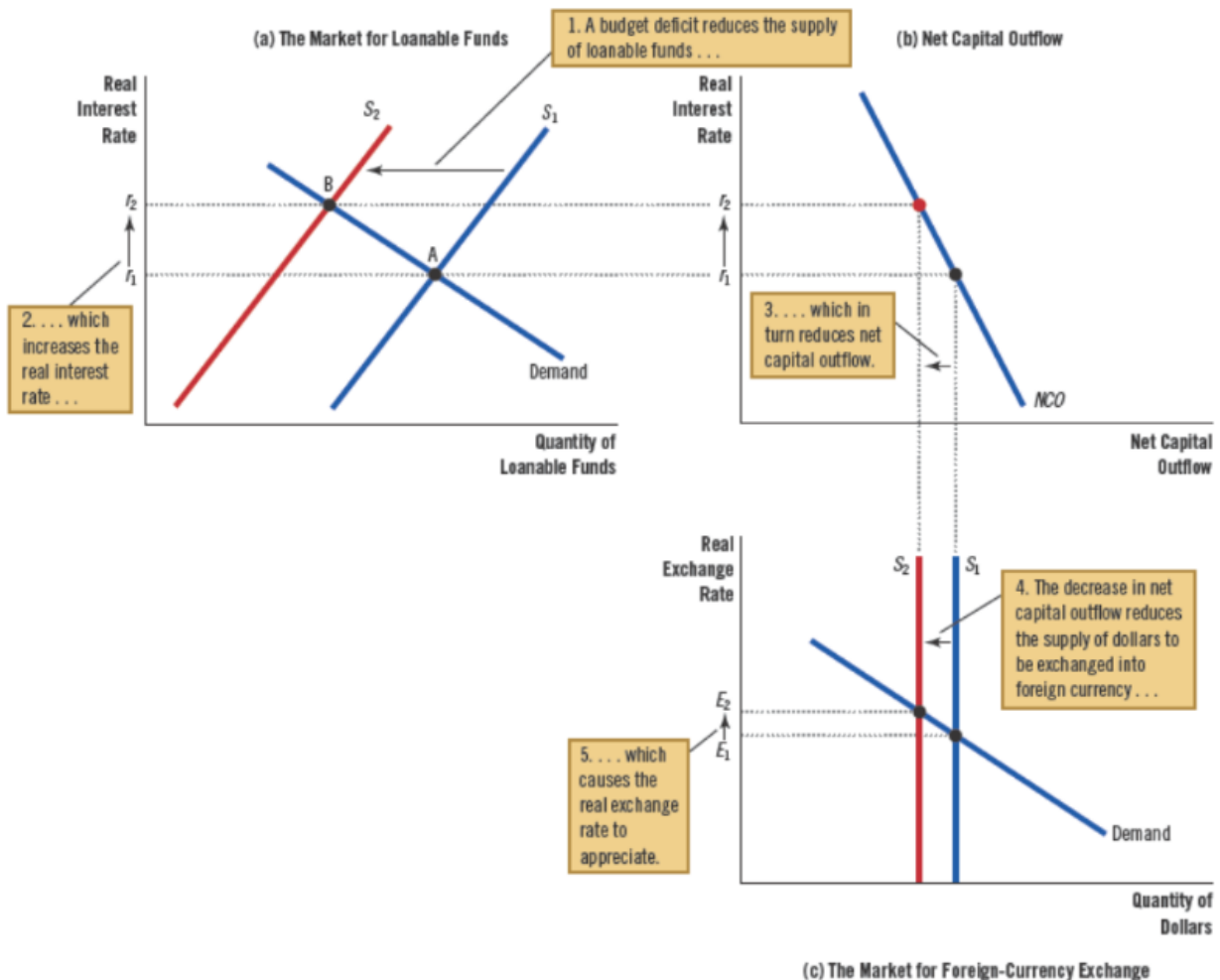
- Simultaneous Equilibrium in Two Markets



- (a) r_1 brings the quantity of loanable funds supplied and demanded into balance
- (b) The interest rate (r_1) comes from panel (a) and determines NCO
- (c) The quantity of NCO from panel (b) determines the supply of dollars to be exchanged into foreign currencies.
 - the real exchange rate does not affect net capital outflow

How Policies and Events Affect an Open Economy

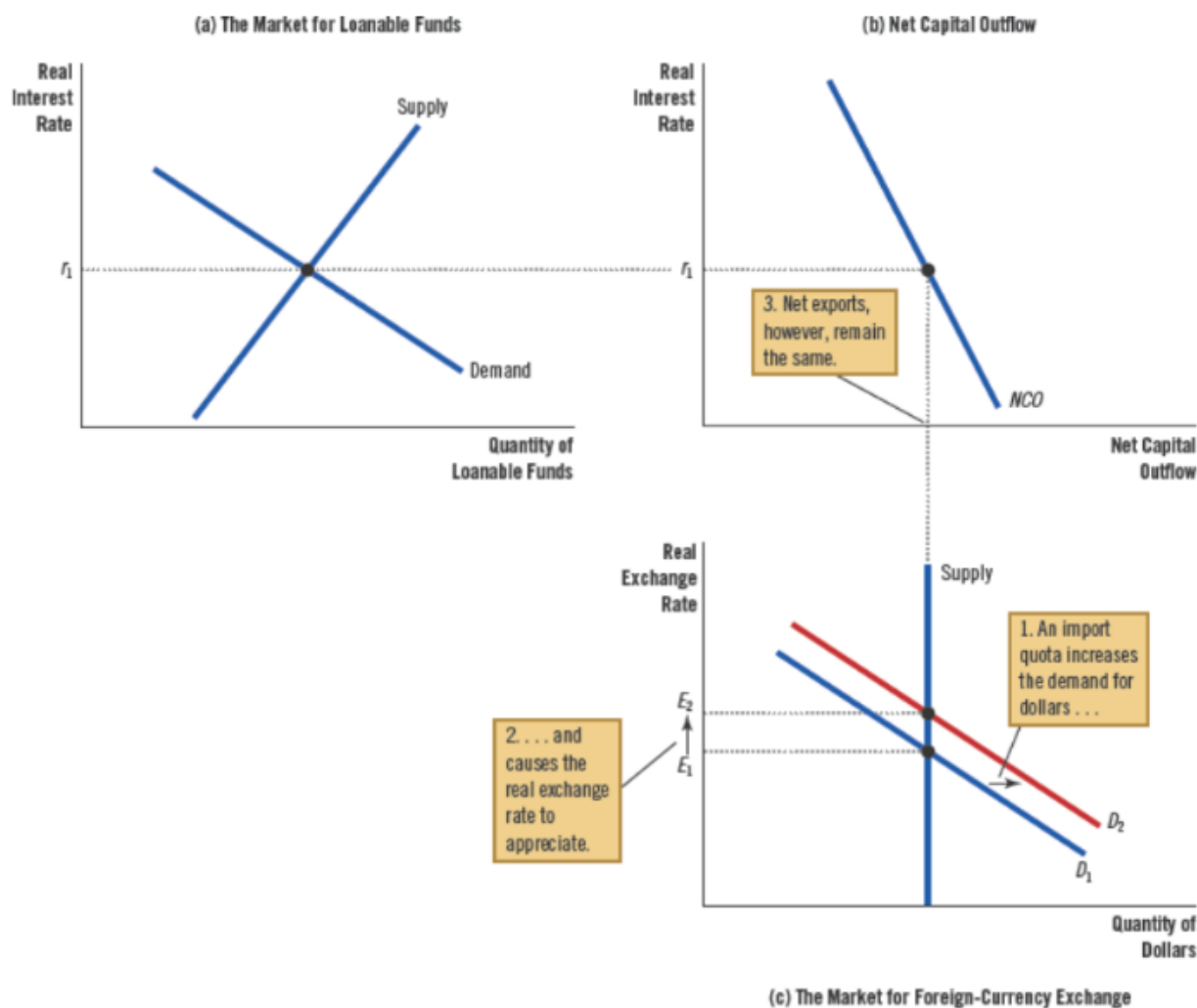
1. Government Budget Deficits



- Budget Deficit ($T < G$): negative public saving
 - it reduces national saving ($S \downarrow$) and shifts the supply curve for loanable funds to the left ($S_1 \Rightarrow S_2$)
 - The interest rate rises from r_1 to r_2 , and this reduces domestic investment (crowd-out effects) and NCO
 - As a result, the supply curve for dollars shifts leftward from S_1 to S_2 and the reduced supply of dollars causes the real exchange to appreciate from E_1 to E_2 . This appreciation makes domestic goods more expensive compared with foreign goods, so the net exports fall ($NX \downarrow$)
- Twin Deficit
 - in an open economy, government **budget deficits** raise real interest rates, crowd out domestic investment, cause the currency to appreciate, and lead to the **trade deficits**

2. Trade Policy

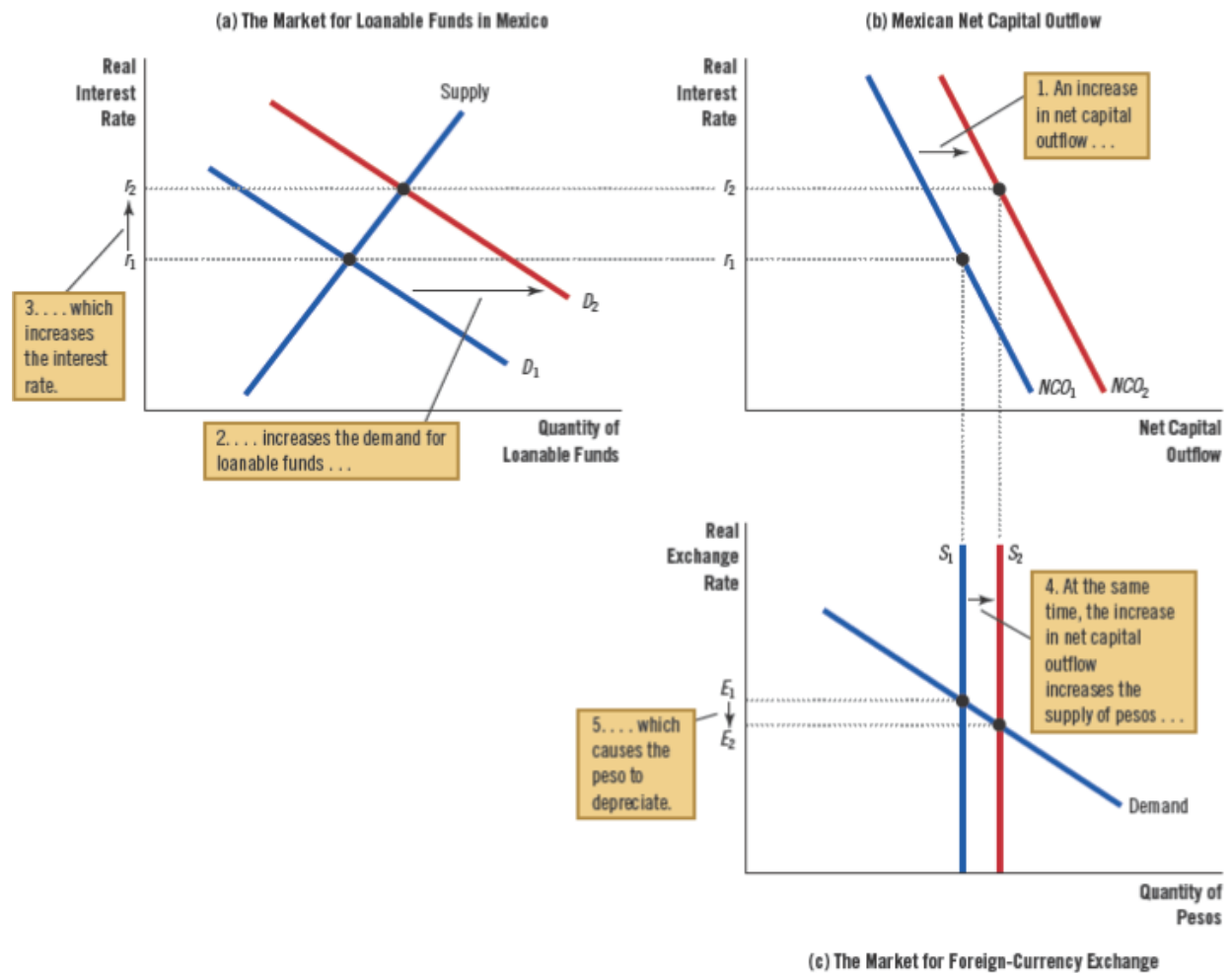
- Trade policy: government policy that directly influences the quantity of goods and services that a country imports or exports
 - Tariff: a tax on imported goods
 - Import quota: a limit on the quantity of a good produced abroad that can be sold domestically



- Import restrictions will reduce imports (M) $\Rightarrow$ then the net exports ($NX = X - M$) will rise for any given real exchange rate
- Because foreigners need dollars to buy U.S. goods, the rise in net exports increases the demand for dollars in the market for foreign-currency, and this shifts the D_1 to the right D_2
- As a result, the real exchange rate appreciates from E_1 to E_2
- Summary: the trade policies do not affect the trade balance

3. Political Instability and Capital Flight

- Capital Flight: a large and sudden reduction in the demand for assets located in a country
 - e.g.) In 1994, Mexico's political instability made the nation seem like a riskier place to invest. People decided to pull some of their assets out of Mexico and move these funds to the United States



- NCO curve shifts to the right from NCO_1 to NCO_2 because of the capital flight
- As a result, the demand for loanable funds shifts from D_1 to D_2 , and this drives up the Mexican real interest rate from r_1 to r_2
- In the market for foreign-currency exchange, the supply of pesos rises from S_1 to S_2 and this depreciates the real exchange rate from E_1 to E_2
- Summary: the capital flight from Mexico increases Mexican interest rates and decreases the value of the Mexican peso in the market for foreign-currency exchange

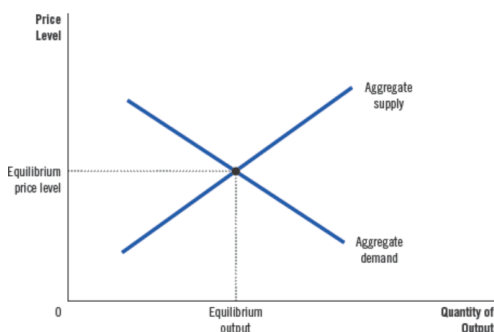
Chapter 20. Aggregate Demand and Aggregate Supply(AD-AS)

Explaining Short-run economic fluctuations

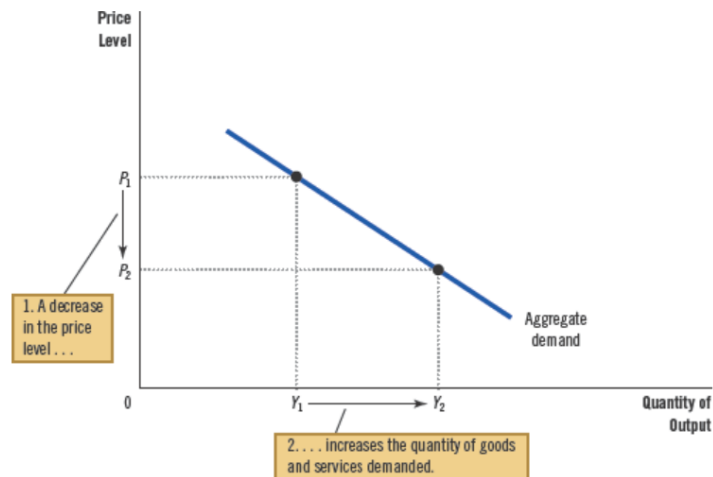
- The assumptions of classical economics
 - (1) Classical dichotomy: the separation of variables into real variables(those that measures quantities or relative prices) and nominal variables(those measured in terms of money)
 - (2) Monetary neutrality: changes in the money supply affect nominal variables but not real variables in the **long-run**.
- The reality of short-run fluctuations
 - In the short-run, real and nominal variables are highly intertwined, and changes in the money supply can temporarily push real GDP away from its long-run trend

The model of Aggregate Demand and Aggregate Supply(AD-AS Model)

- AD-AS Model
 - the model that most economists use to explain short-run fluctuations in economic activity around its long-run trend
- (1) Aggregate-demand curve(AD)
 - A curve that shows the quantity of goods and services that households(C), firms(I), the government(G), and customers abroad want to buy(NX) at each price level
- (2) Aggregate-supply curve(AS)
 - A curve that shows the quantity of goods and services that firms choose to produce and sell(Y) at each price level



The Aggregate-Demand Curve(AD)



- Why the AD curve slopes downward

- (1) The Wealth Effect(between P and C)

- A decrease in P raises the real value of money and makes consumers wealthier, encouraging them to spend more
 $\Rightarrow$ The increase in C means a larger quantity of goods and services demanded
- Conversely, an increase in P reduces the real value of money and makes consumers poorer
 $\Rightarrow$ reducing C and the quantity of goods and services demanded

- (2) The Interest-Rate Effect(between P and I)

- A lower P induces people do not need to hold as much money to buy the goods and services they want
 $\Rightarrow$ People try to reduce their money holdings by lending some of it out, and this drives down interest rates(r)
 $\Rightarrow$ Reduced r encourages spending on I and increases the quantity of goods and services demanded
- Conversely, a higher P reduces the r
 $\Rightarrow$ discourages I and decreases the quantity of goods and services demanded

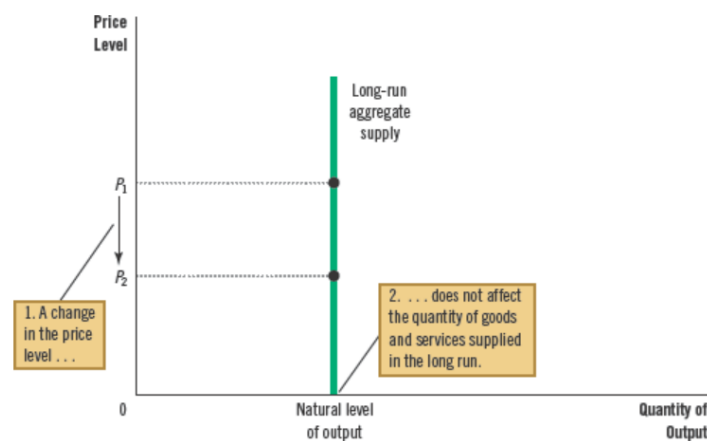
- (3) The Exchange-Rate Effect(between P and NX)

- Real exchange rate equation: $RER = \frac{e*P}{P^*}$
- If the U.S. price(P) falls, RER decreases
 $\Rightarrow$ this depreciation stimulates the U.S. NX and increases the quantity of goods and services demanded
- Conversely, when the U.S. price(P) rises, RER increases
 $\Rightarrow$ this appreciation reduces the U.S. NX and the quantity of goods and services demanded

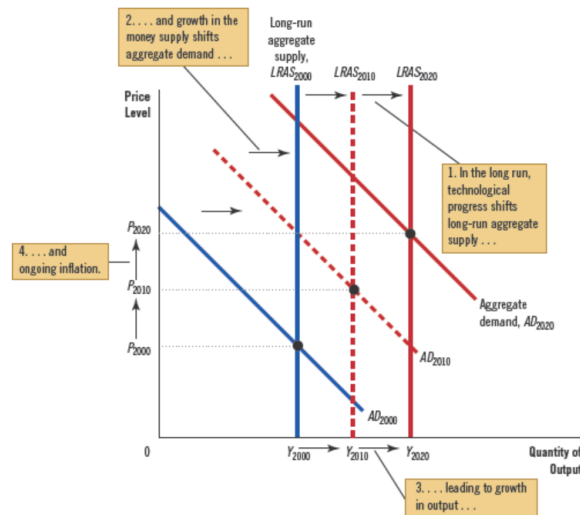
- Why the AD curve might shift
 - (1) Shift arising from changes in C
 - Stock market boom
 - Preferences for consumption today
 - Tax cuts
 - (2) Shift arising from changes in I
 - Better macroeconomic expectations, optimism
 - Lower interest rates, expansionary policy
 - Investment tax credit or other tax incentives
 - (3) Shift arising from changes in G
 - Increase in Federal spending
 - Increase in State and local spending
 - (4) Shift arising from changes in NX
 - Booms in foreign countries that buy the domestic country's exports
 - Depreciation of domestic currency resulting from international speculation in the foreign exchange market

The Aggregate-Supply Curve(AS)

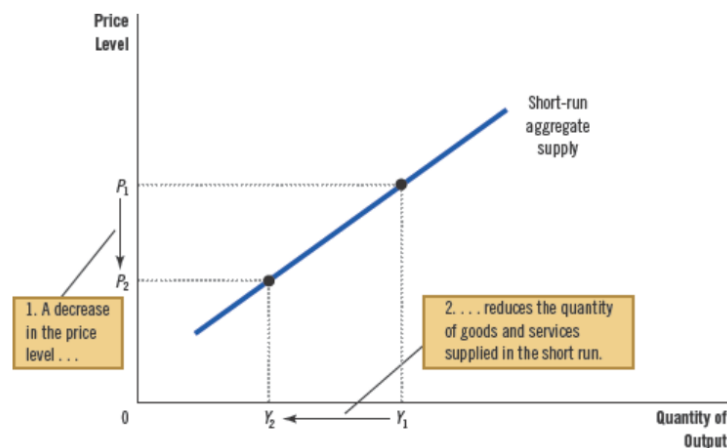
- Long-run aggregate-supply curve($LRAS$)
 - In the long-run, the economy's labor, capital, natural resources, and technology determine the total quantity of goods and services supplied, and **this quantity supplied is the same regardless of the price level**
 - * Natural rate of output: the production of goods and services that an economy achieves in the long run when unemployment is at its normal rate
 - The vertical $LRAS$ curve is a graphical representation of the classical dichotomy and monetary neutrality



- Why the *LRAS* might shift
 - $Y = A \cdot F(L, K, H, N)$
 - Increase in L or decrease in natural rate of unemployment
 - Increase in physical capital K or human capital H
 - Discovery of new natural resources N
 - Technological improvements A
- Long-run growth and inflation using *AD* and *LRAS*



- Why the *SRAS* slopes upward
 - By contrast, in the short-run, the price level does affect the economy's output
 - * In the short-run, an increase in the price level tends to increase the quantity of goods and services supplied
 - * Conversely, a decrease in the price level tends to reduce the quantity of goods and services supplied



- Three Theories

$$Y = Y_N + \alpha(P - P_E)$$

Y_N : Natural rate of output(in the long-run)

P_E : Expected price level

(1) The Sticky-Wage Theory

- Nominal wages are slow to adjust to changing economic conditions. Wages are "sticky" in the short-run
- e.g.) Suppose the price level changes from $100(P_E)$ to $105(P)$ and the nominal wage remains stuck at \$30. In response, it hires more workers and increase the quantity of output supplied

(2) The Sticky-Price Theory

- The slow adjustment of the prices of some goods and services occurs in part because there are costs to changing prices, called **menu costs**
- e.g.) Suppose that each firm in the economy announces its prices in advance based on the economic conditions it expects for the coming year(P_E). But the actual price level(P) turns out to be higher($P > P_E$) than expected prices. Because of the menu costs, in the short-run, some firms will not raise their price instantly, and the relatively low prices attract customers, inducing these firms to increase employment and production.

(3) The Misperception Theory

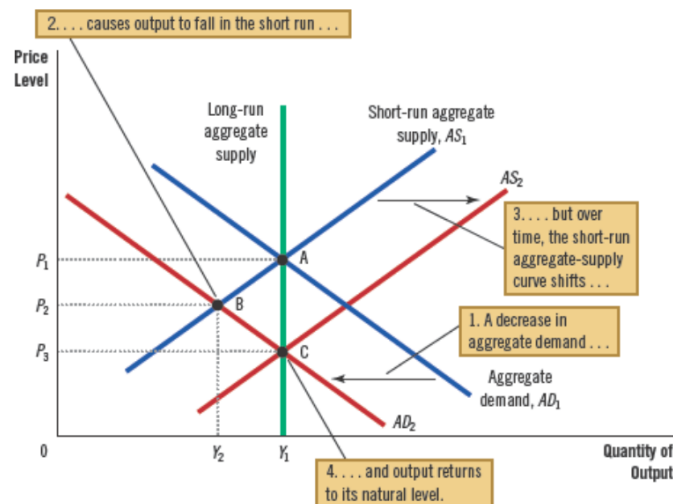
- It holds that changes in the overall price level can temporarily mislead suppliers about what is happening in the markets in which they sell their output.
- e.g.) When $P > P_E$, suppliers of goods and services may notice the price of their output rising and infer, *mistakenly*, that their *relative* prices are rising. They would conclude that it is a good time to produce more.

- Why the *SRAS* might shift

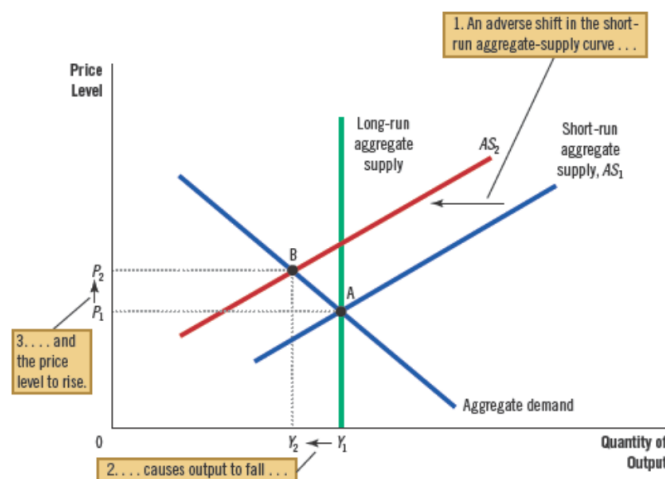
- An increase in the P_E reduces the quantity of goods and services supplied and shifts the *SRAS* to the left.
- A decrease in the P_E raises the quantity of goods and services supplied and shift the *SRAS* to the right

Two Causes of Economic Fluctuations

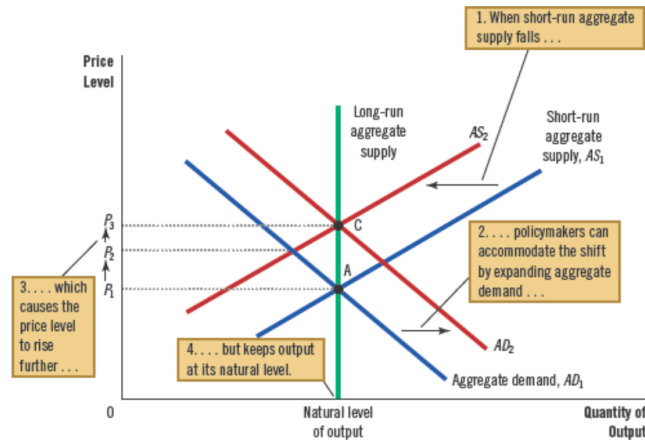
- The effects of a shift in AD
 - e.g.) If a wave of pessimism overtakes the economy.
 - * $A \Rightarrow B$: The pessimism affects spending plans, so AD curve will shift to the left. As a result, the price level falls from P_1 to P_2 and output falls from Y_1 to Y_2
 - * $B \Rightarrow C$: P_E falls because of the movement from A to B , it shifts the AS to the right



- The effects of a shift in AS
 - e.g.) Suppose that the costs of production suddenly increase (bad weather, environmental issues...)
 - * $A \Rightarrow B$: Because production costs affect the firms that supply goods and services, AS curve shifts to the left. As a result, the output falls and the price level rises (stagflation)
 - * $B \Rightarrow A$: if the AD is held constant, AS curve shifts to the right again and returns to the original point A.



- * $A \Rightarrow C$: However, in the real world, monetary and fiscal policymakers might attempt to offset the effect of $A \Rightarrow B$ by shifting the AD curve to the right.
 - Because of the policy, the output will remain at its natural level, but price levels rises from P_1 to P_3



Chapter 21. The Influence of Monetary and Fiscal Policy on Aggregate Demand

How Monetary Policy Influences Aggregate Demand

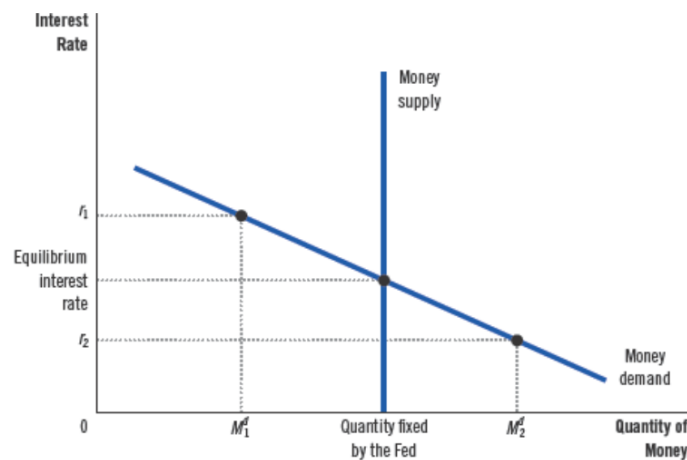
- Three reasons for the AD curve's downward slope(from chapter 20)
 - (1) The Wealth Effect
 - (2) The Interest-rate Effect
 - (3) The Exchange-rate Effect
- For the U.S. economy, the most important reason for the downward slope of the AD curve is the interest-rate effect
 - Money holdings and trade represent only a small fraction of U.S. GDP. Therefore, the wealth effect and the exchange-rate effect are relatively small in the U.S. economy
- **Theory of liquidity preference**
 - Keynes's theory that the interest rate adjusts to bring money supply and money demand into balance
 - In the short run, we assume that the expected rate of inflation(π^e) is constant
$$r = i - \pi^e$$
 - * In this case, nominal(i) and real(r) interest rates differ by a constant(π^e)
 - * When the i rises or falls, the r rises or falls by the same amount because the π^e is constant in the short-run
 - Money Supply
 - * the Federal Reserve control the money supply in the U.S.(ex. open-market operation, reserve requirements, discount rate, federal funds rate...)
 - * Because the quantity of money supplied in the economy is fixed at whatever level the Fed decides to set it, it does not depend on the prevailing interest rate(vertical money supply curve)

– Money Demand

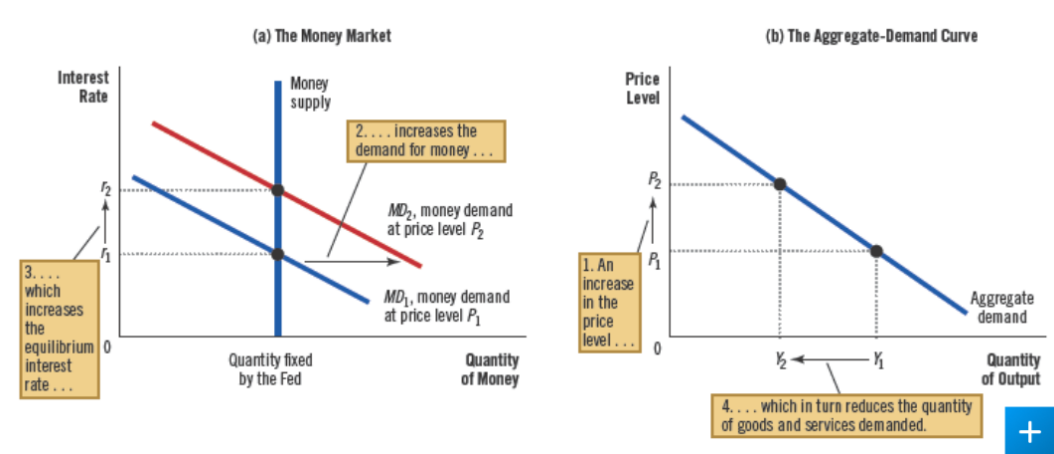
- * **Liquidity** refers to the ease with which it can be converted into the economy's medium of exchange
- * Liquidity explains why there is always some demand for money
 - People choose to hold money instead of other assets that offer higher rates of return because they can use it to buy goods and services
- * interest rate: it represents the **opportunity cost** of holding money (price of holding money)
 - e.g.) When you hold wealth as cash rather than as an interest-bearing bond or bank deposit, you lose the interest you could have earned
 - Thus, the money demand curve has a negative slope (downward sloping below)

– Equilibrium in the money market

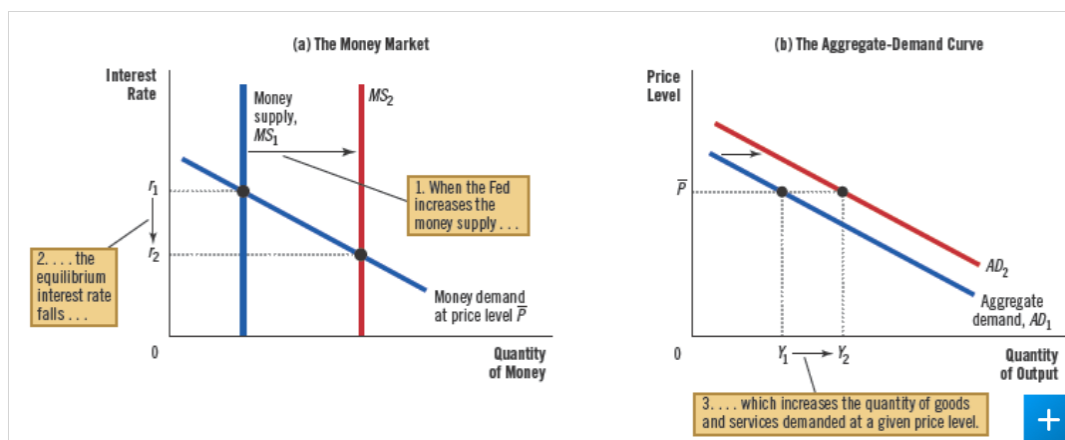
- * According to the theory of liquidity preference, the interest rate adjusts to balance the supply and demand for money



- The Downward slope of the AD curve
 - The price level is one determinant of the quantity of money demand
 - * At higher prices, more money is needed to buy a good or service every time, so people will choose to hold a larger quantity of money ($P \uparrow \Rightarrow$ money demand $\uparrow$)
 - Three steps of understanding why the AD curve has downward slope
 - (1) A higher price level raises money demand (shift the money demand curve to the right)
 - (2) Higher money demand leads to a higher interest rate
 - (3) A higher interest rate reduces the quantity of goods and services demanded

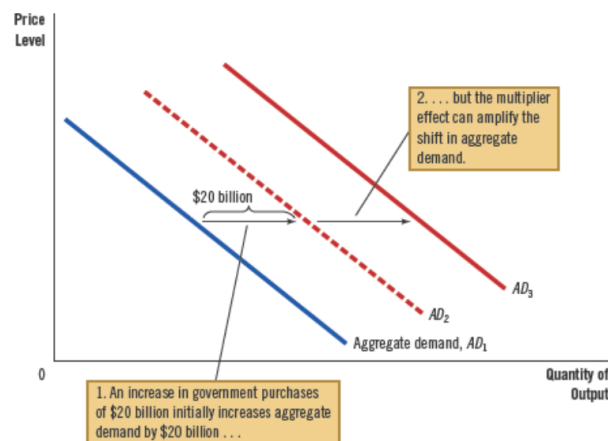


- Changes in the Money Supply
 - If the Fed increase the money supply (monetary injection), then the money supply curve shifts to the right
 - Because the money demand curve has not changed, the interest rate falls
 - When the interest rate falls, it induces people to hold the additional money, thus households and firms will spend more money on consumption and investment. As a result, AD curve shift to the right



How Fiscal Policy Influences AD

- Fiscal Policy
 - the setting of the levels of government spending and taxation by government policymakers
- Changes in Government Purchases(G)
 - When the government alters its own purchases of goods and services, it shifts the AD curve directly($AD = C + I + G + NX$)
 - There are two macroeconomic effects that cause the size of the shift in AD to differ from the change in government purchases(the multiplier effect, and the crowding-out effect)
- The Multiplier Effect
 - the additional shifts in AD that result when expansionary fiscal policy increases income and thereby increases consumer spending
 - e.g.) When the government purchases an airplane from Boeing, it raises employment and profits at Boeing. Then, owners and workers respond to this increase in income by raising their own spending on consumer goods.
 - Once all these effects are added together, the total impact on the quantity of goods and services demanded can be much larger than the initial boost from higher government spending



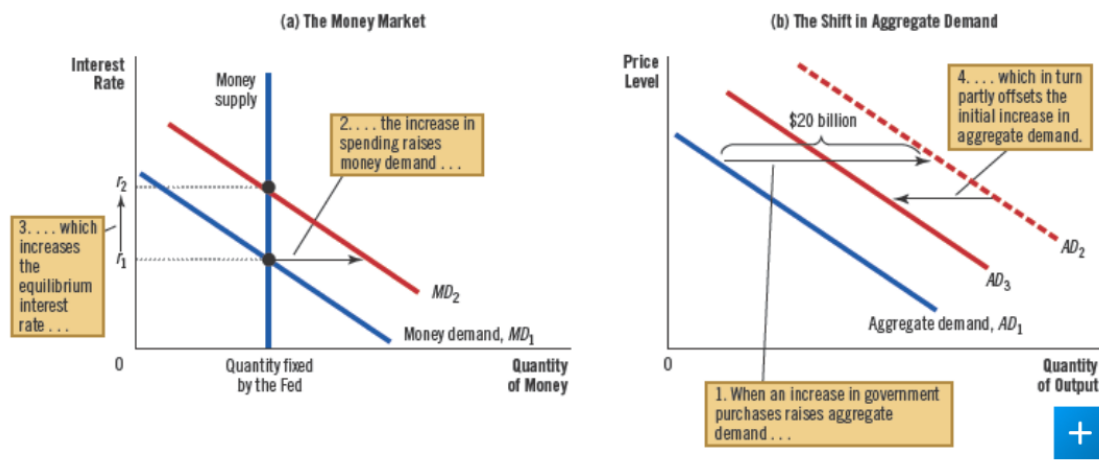
- A Formula for the Spending Multiplier

$$\text{Multiplier} = \frac{1}{(1 - MPC)}$$

- Marginal propensity to consume(MPC): the fraction of extra income that a household consumes rather than saves
- e.g.) If G is \$10 billion, and MPC is 20%
 - $\Rightarrow \text{multiplier} = \frac{1}{(1-20\%)} = \frac{1}{(1-1/5)} = \frac{5}{4}$.
 - $\Rightarrow$ Thus, the effect will be $\$10 \text{ billion} \times \frac{5}{4} = \12.5 billion

• The Crowding-out Effect

- the offset in AD that results when expansionary fiscal policy raises the interest rate and thereby reduces investment spending
- While an increase in G stimulates the AD for goods and services (by multiplier effect), it also causes the interest rate to rise, reducing investment spending and putting downward pressure on AD
- e.g.) When the government buys planes from Boeing, it raises the incomes of the workers and owners $\Rightarrow$ As income rises, households plan to buy more goods and services and so **choose to hold more of their wealth in liquid form** $\Rightarrow$ Money demand curve shifts to the right $\Rightarrow$ the interest rate goes up $\Rightarrow$ Investment will decrease
- This crowding-out effect partially offsets the impact of government purchases on AD



Reference

1. Mankiw, N. Gregory. *Principles of Macroeconomics*. 10th ed. Cengage Learning, 2024.